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(54) SEMICONDUCTOR DEVICE

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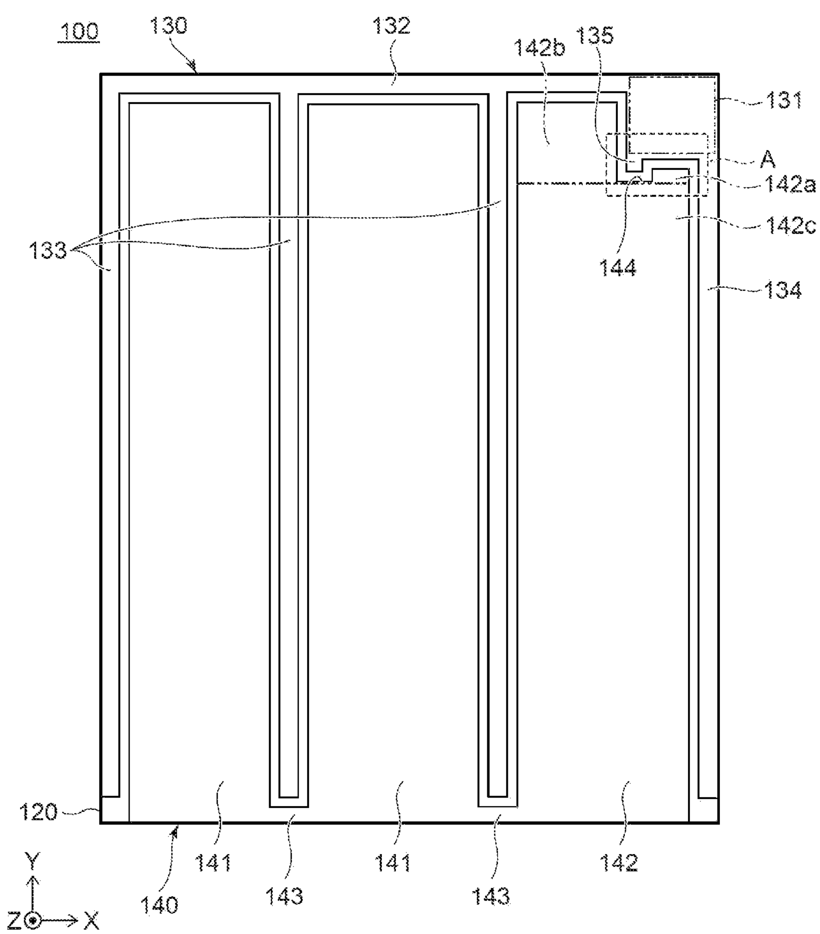
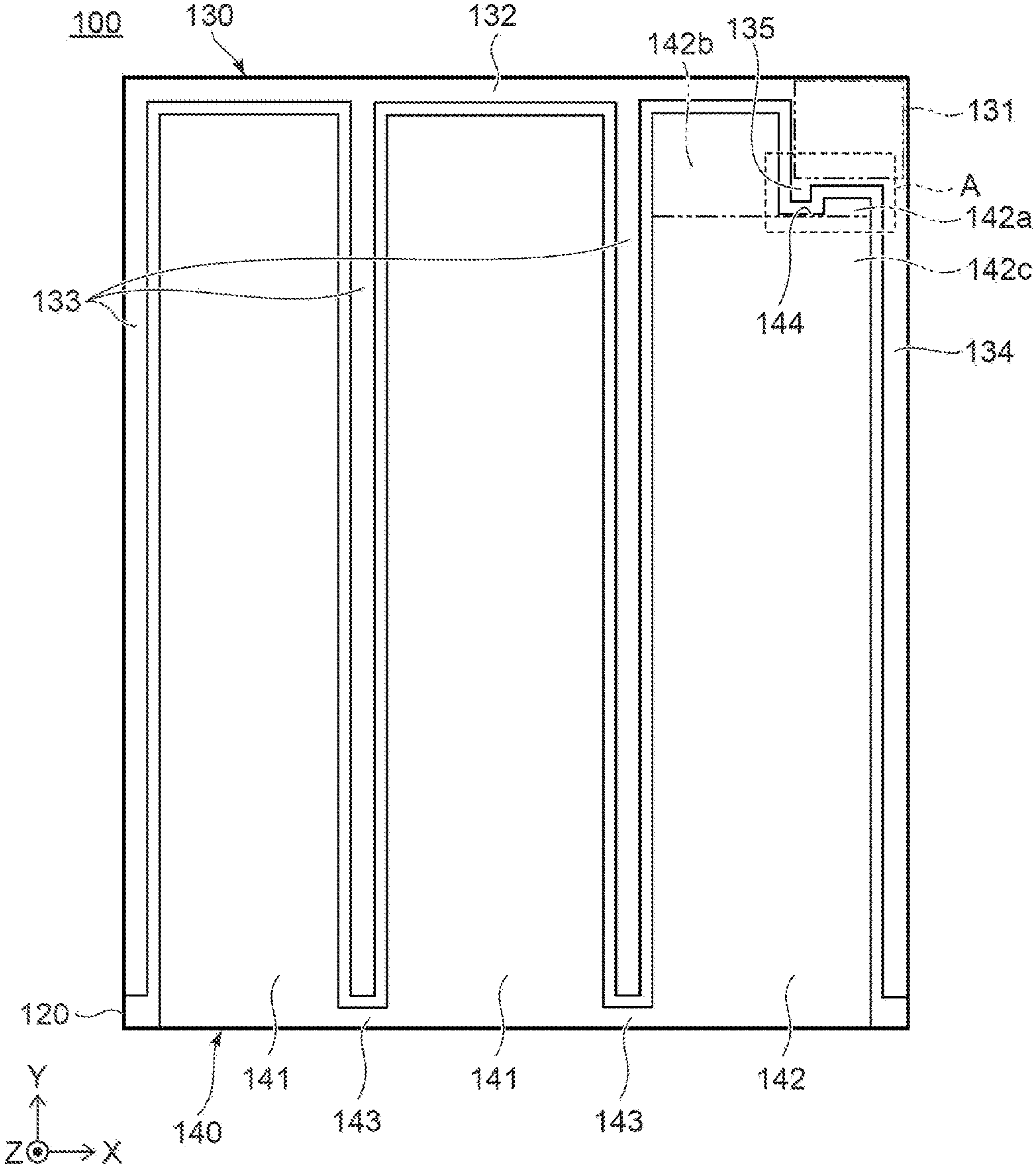
(57) **ABSTRACT**

A semiconductor device includes: a second electrode, located in a semiconductor part, extending in a first direction; a third electrode, located in the semiconductor part, including a first portion, a second portion, and a first middle portion positioned below the second electrode between the first portion and the second portion, the second electrode being located between the first portion and the second portion in the first direction; a fourth electrode, located above the semiconductor part, including a pad portion separated from the second electrode and the second portion in a second direction, and a protrusion protruding from the pad portion and covering the second electrode and being connected to the second electrode; and a fifth electrode, located above the semiconductor part, including a first covering portion being connected to the first contact portion and a second covering portion being connected to the first portion.

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132
142b
135
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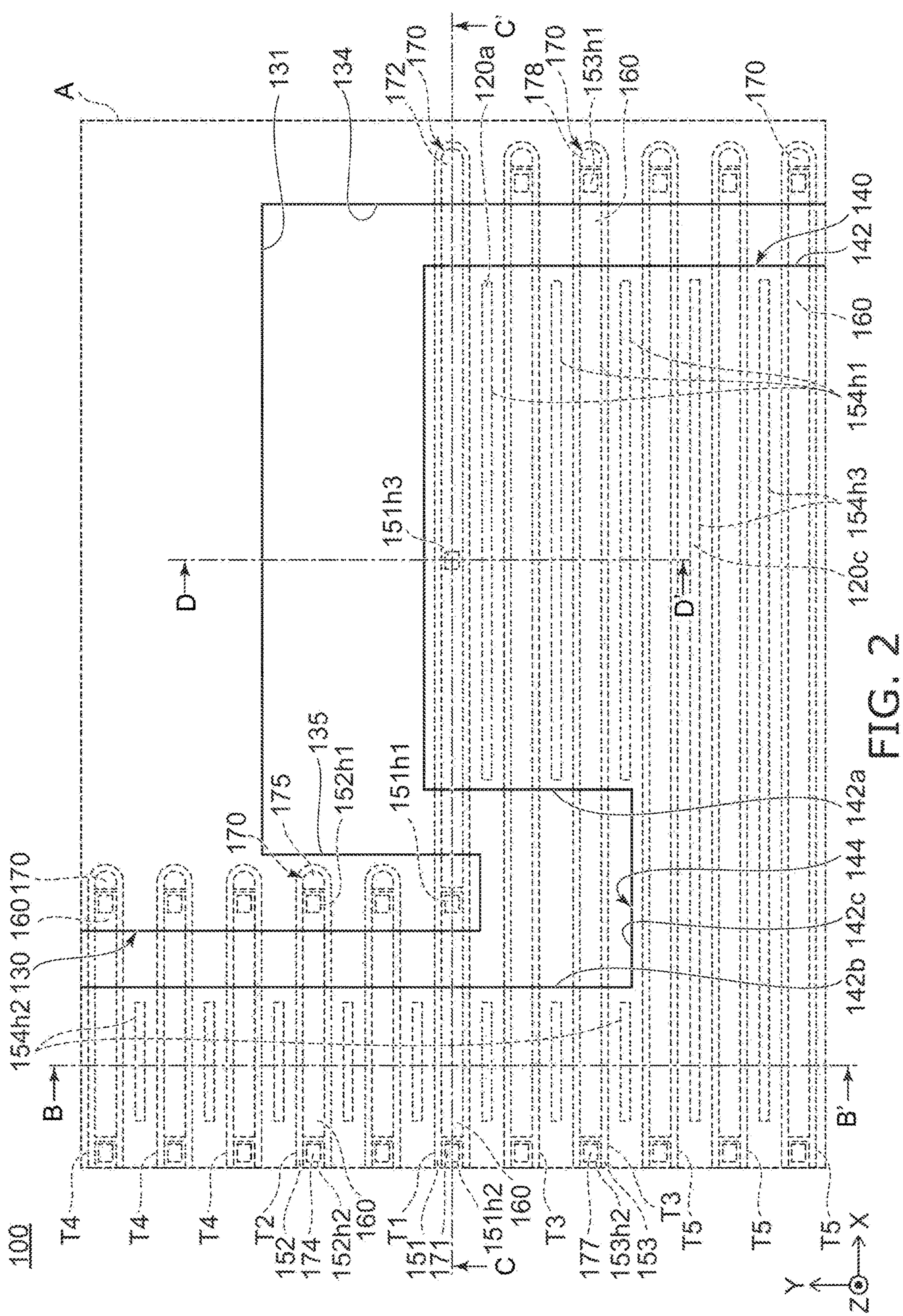
Y
Z
X

FIG. 1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C'
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160
140
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151h3
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D'
135
152h1
175
151h1
142a
144
142c
142b
130
154h2
B
B'
T4
T2
152
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171
151h2
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153h2
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T5
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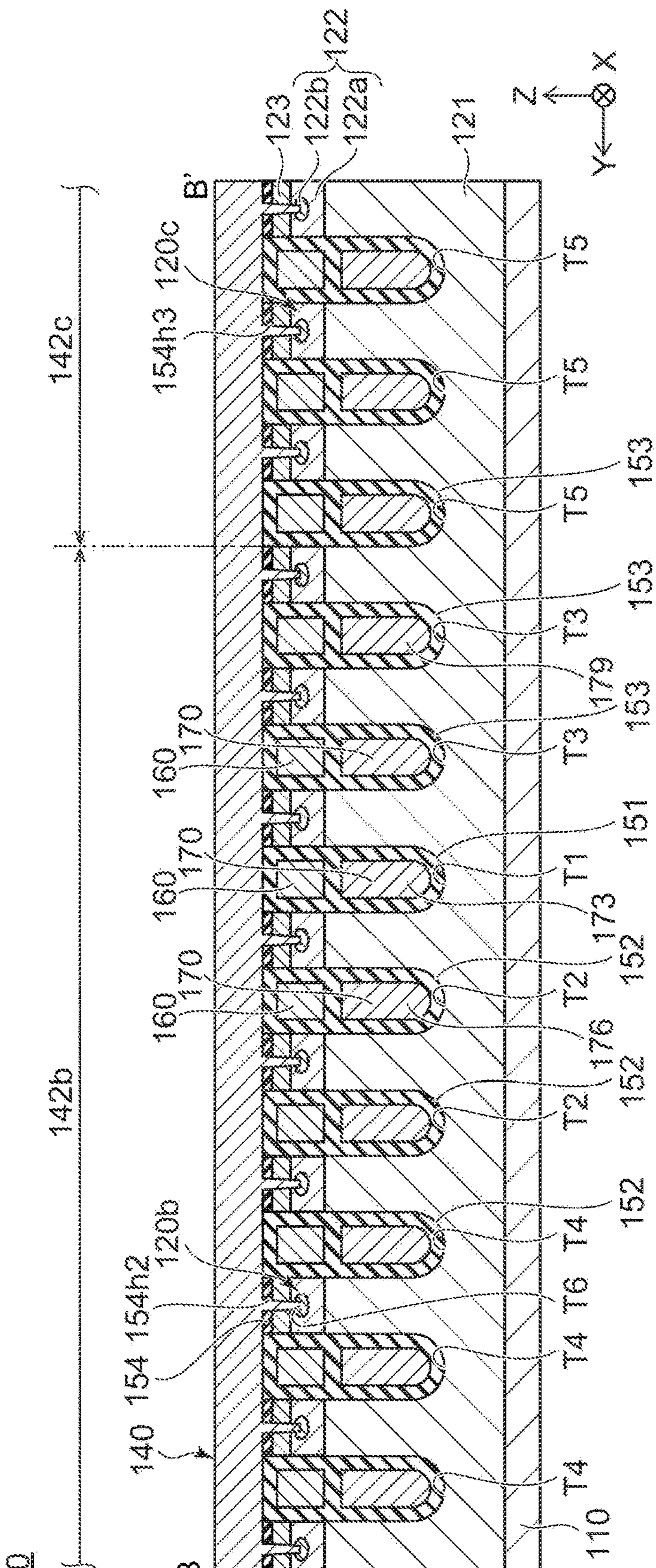
FIG. 2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142b
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B
B'
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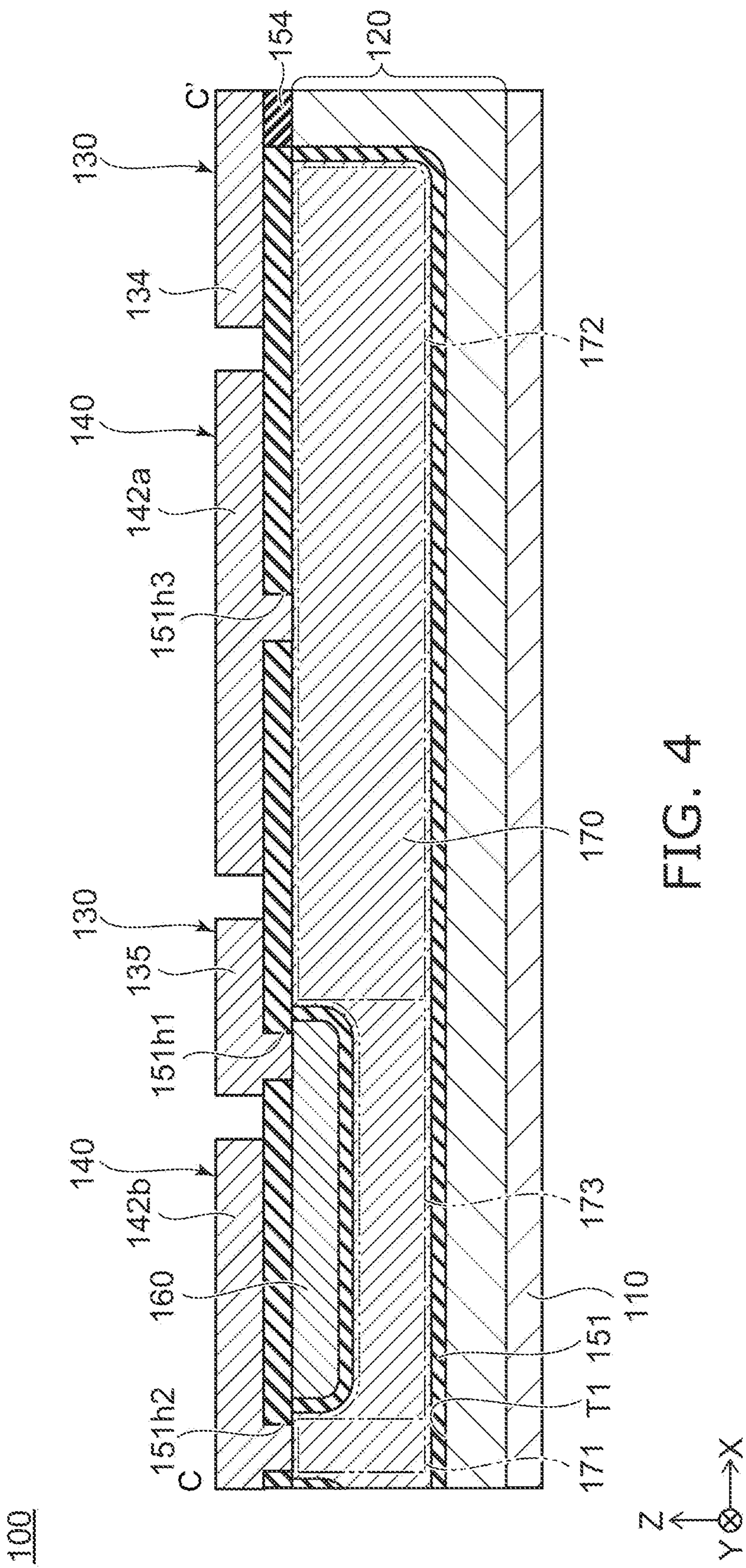
122
122a
122b
123
120b
120c
154
154h2
154h3
151
152
153
160
170
173
176
179
T1
T2
T3
T4
T5
T6
Z
Y
X

FIG. 3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C'
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134
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142a
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142b
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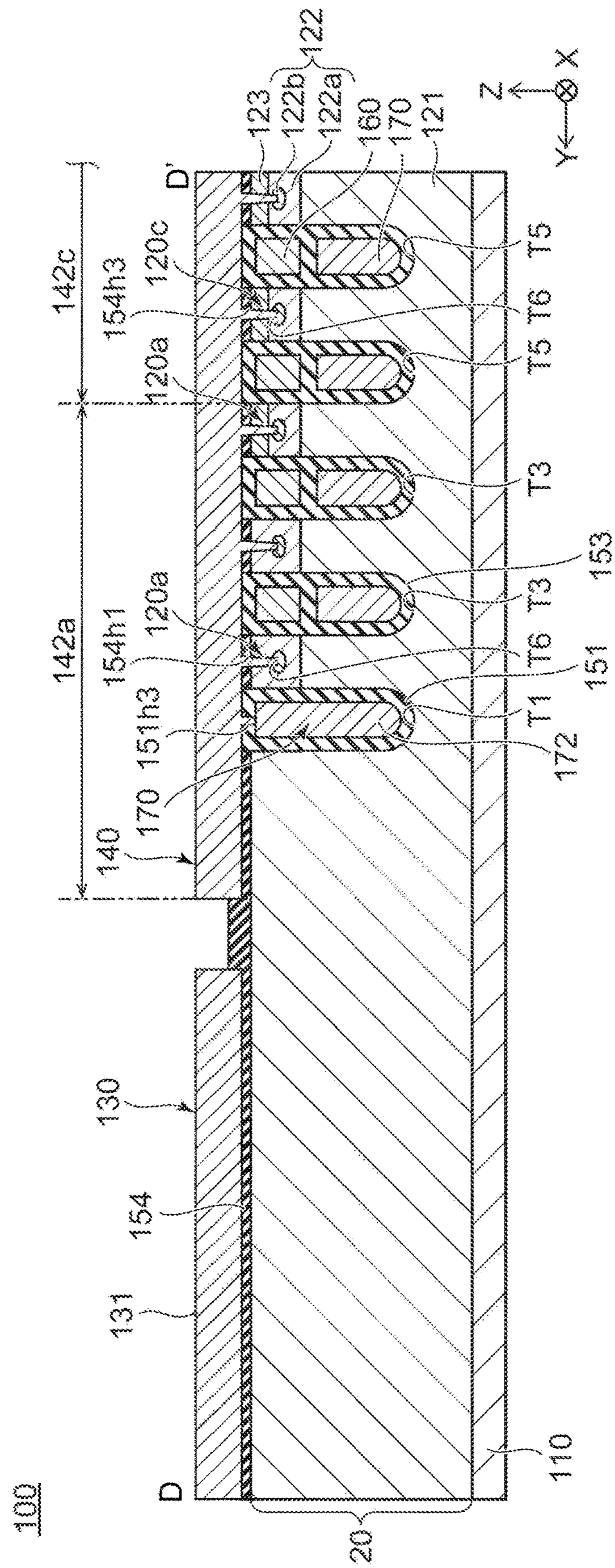
151h2
173
110
151
T1
171
Z
Y
X

FIG. 4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D
D'
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122
122a
122b
123
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131
140
142a
142c
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151h3
153
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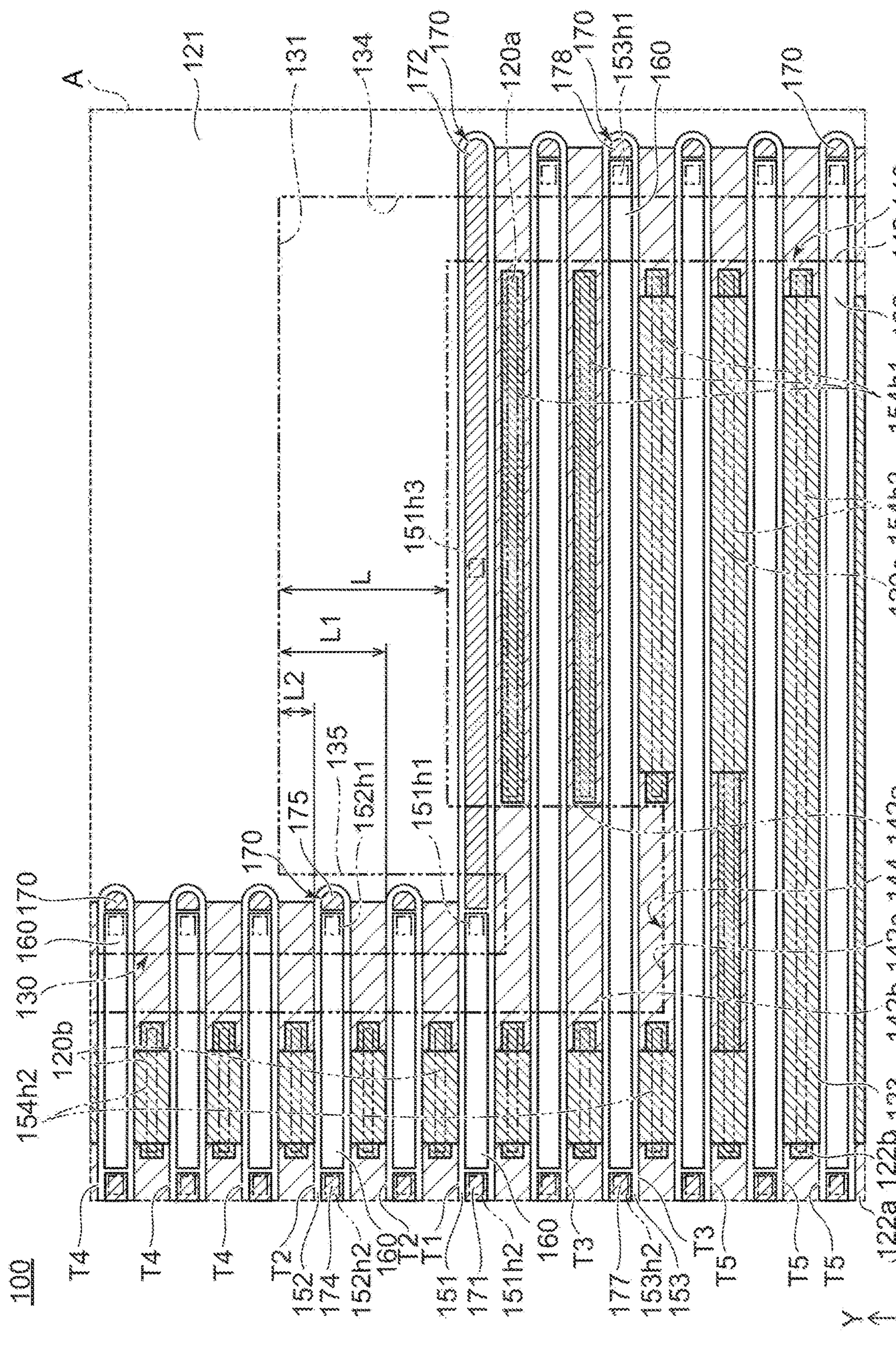
154h1
154h3
160
170
172
120a
120c
T1
T3
T5
T6
Z
X
Y

FIG. 5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154h1
151h3
154h3
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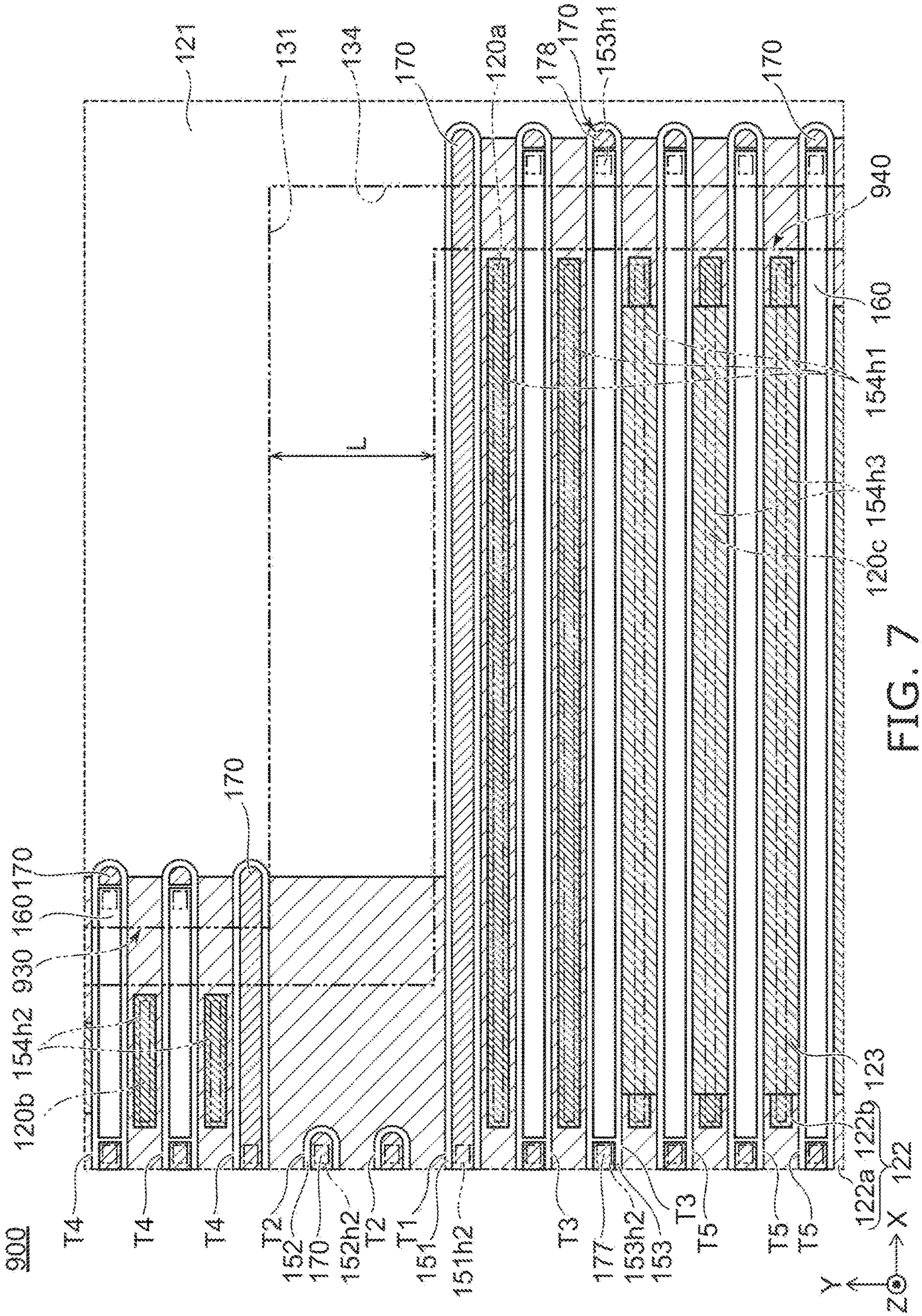
L1
L2
135
175
152h1
151h1
142a
144
142c
142b
123
122b
122a
122
130
120b
154h2
T4
T2
T1
T3
T5
152
174
152h2
151
171
151h2
177
153h2
153
Y
Z
X

FIG. 6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120a
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154h1
154h3
120c
L
120b
154h2
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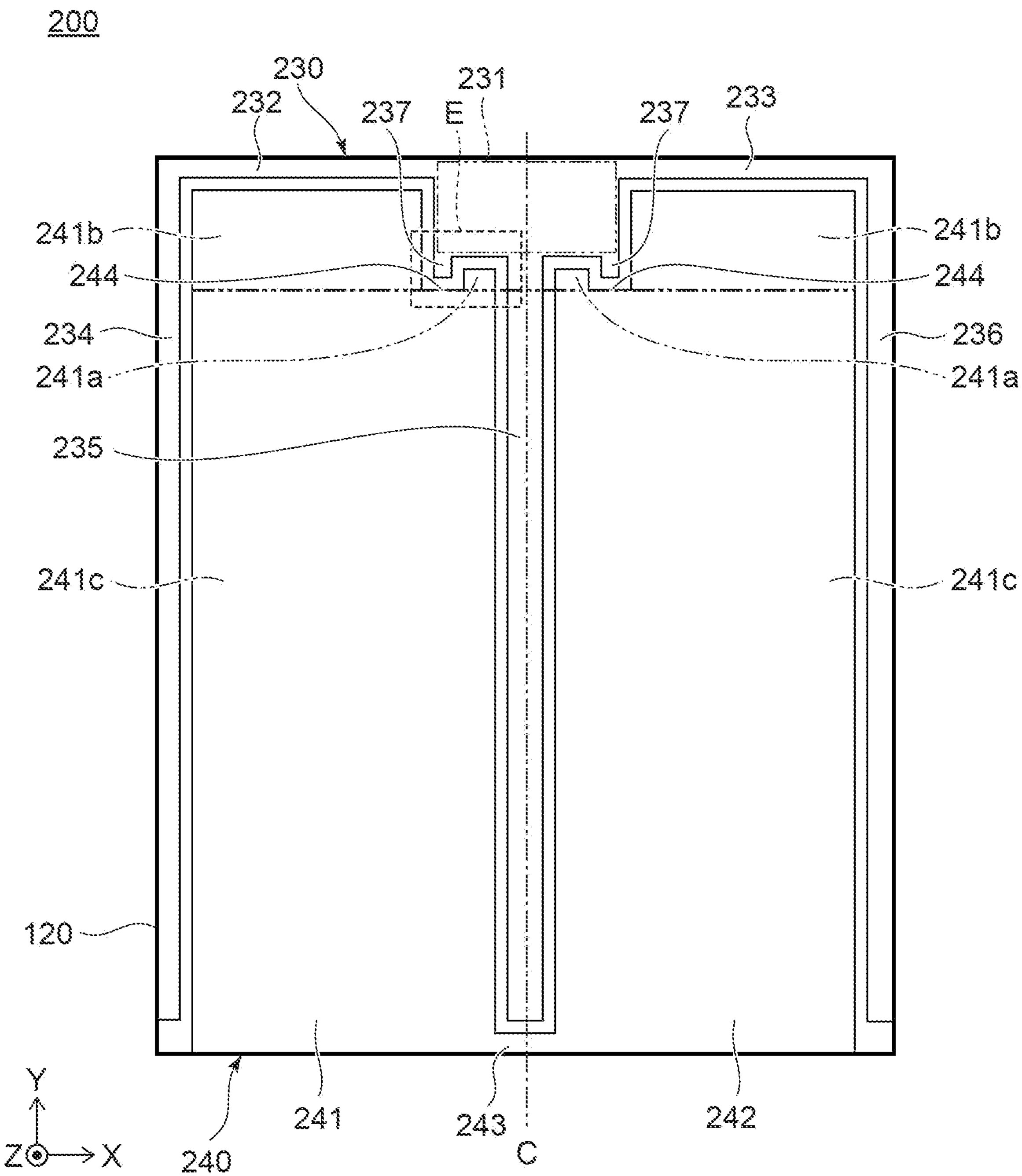
T2
152
152h2
T1
151
151h2
T3
177
153h2
153
T5
123
122a
122b
122
X
Y
Z

FIG. 7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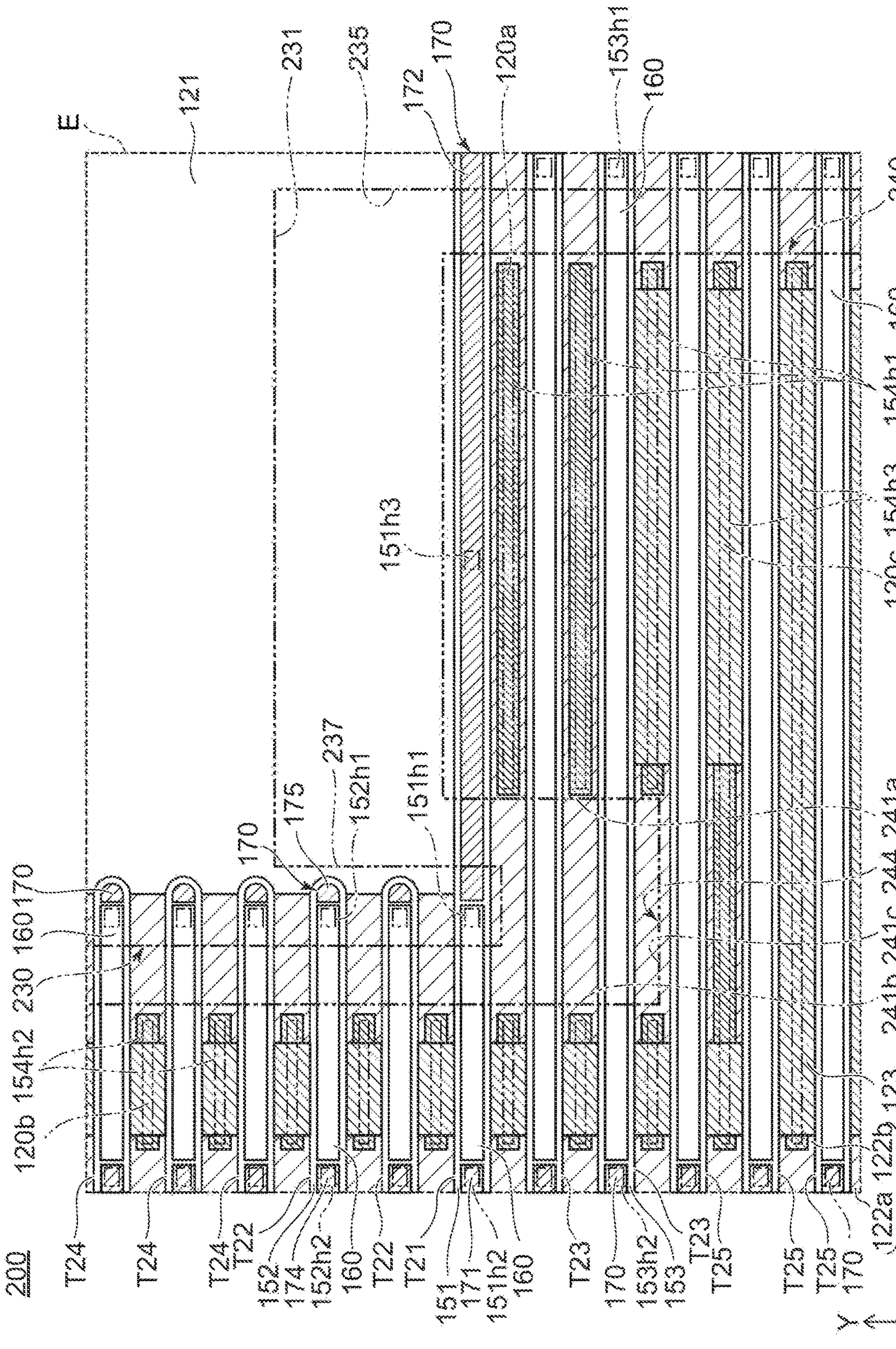
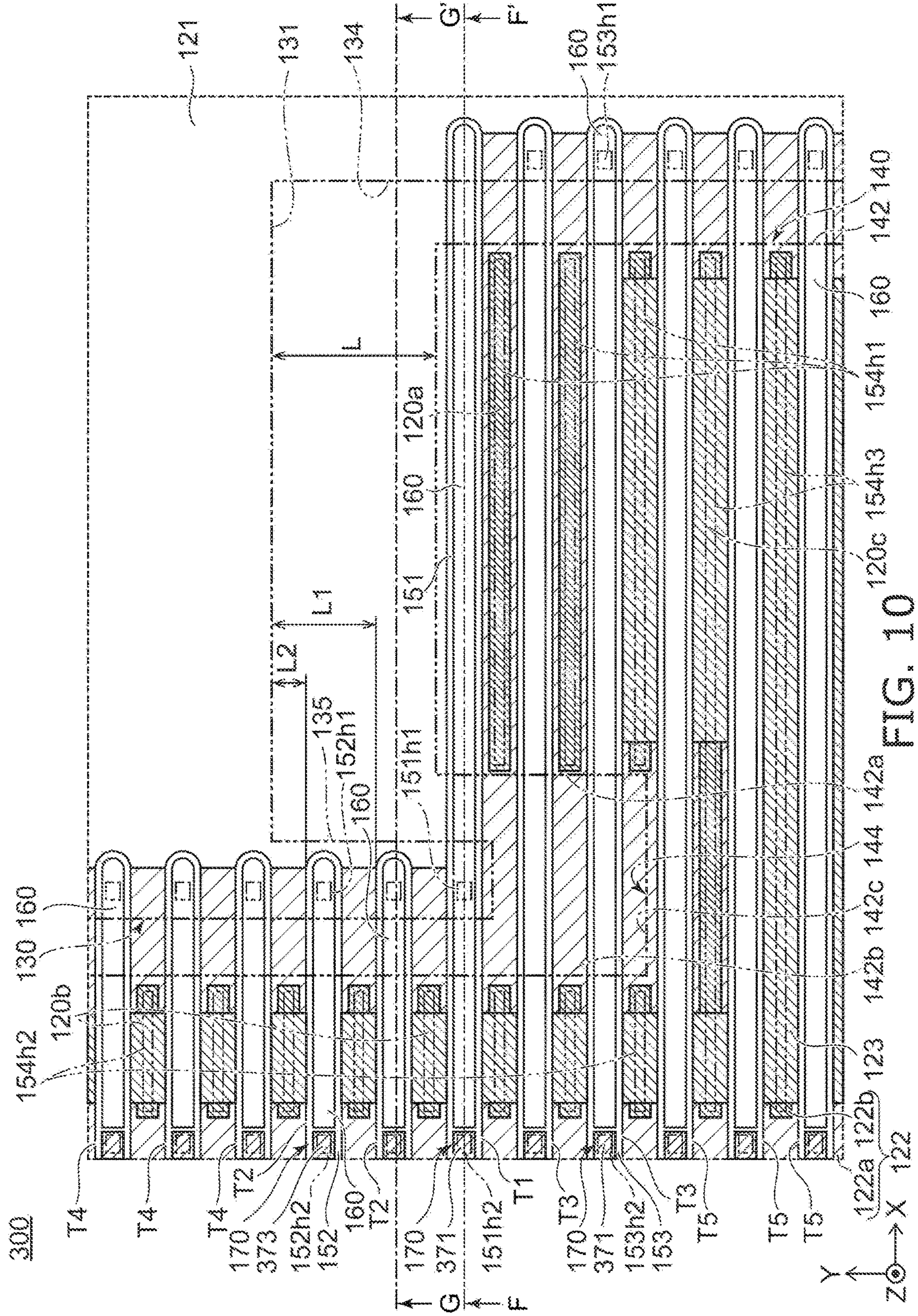
230
231
232
233
237
E
237
241b
241b
244
244
234
236
241a
241a
235
241c
241c
120
Y
Z
X
240
241
243
C
242

FIG. 8



200
E
121
231
235
172
170
120a
153h1
160
240
154h1
154h3
120c
151h3
237
152h1
175
151h1
120b
154h2
230
160
170
241a
244
241c
241b
123
122b
122a
122
T24
T22
T21
T23
T25
152
174
152h2
151
171
151h2
153h2
153
X
Y
Z

FIG. 9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F'
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154h1
154h3
120c
120a
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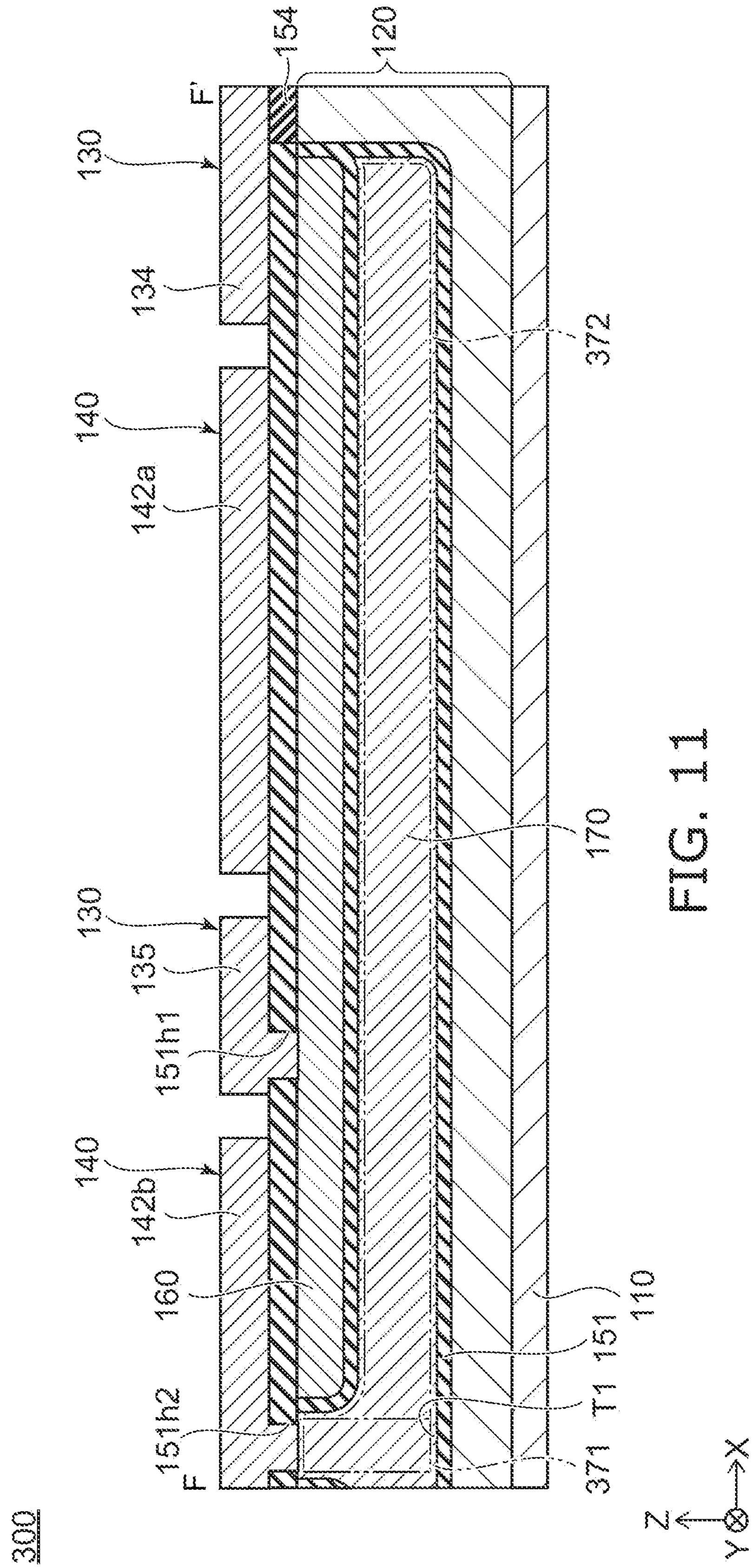
L1
L2
135
152h1
160
151h1
142a
144
142c
142b
123
122b
122a
122
130
160
120b
154h2
T4
T4
T4
T2
170
373
152h2
152
160
T2
170
371
151h2
T1
T3
170
371
153h2
153
T3
T5
T5
T5
G
F
X
Y
Z

FIG. 10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F
F'
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134
140
142a
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135
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142b
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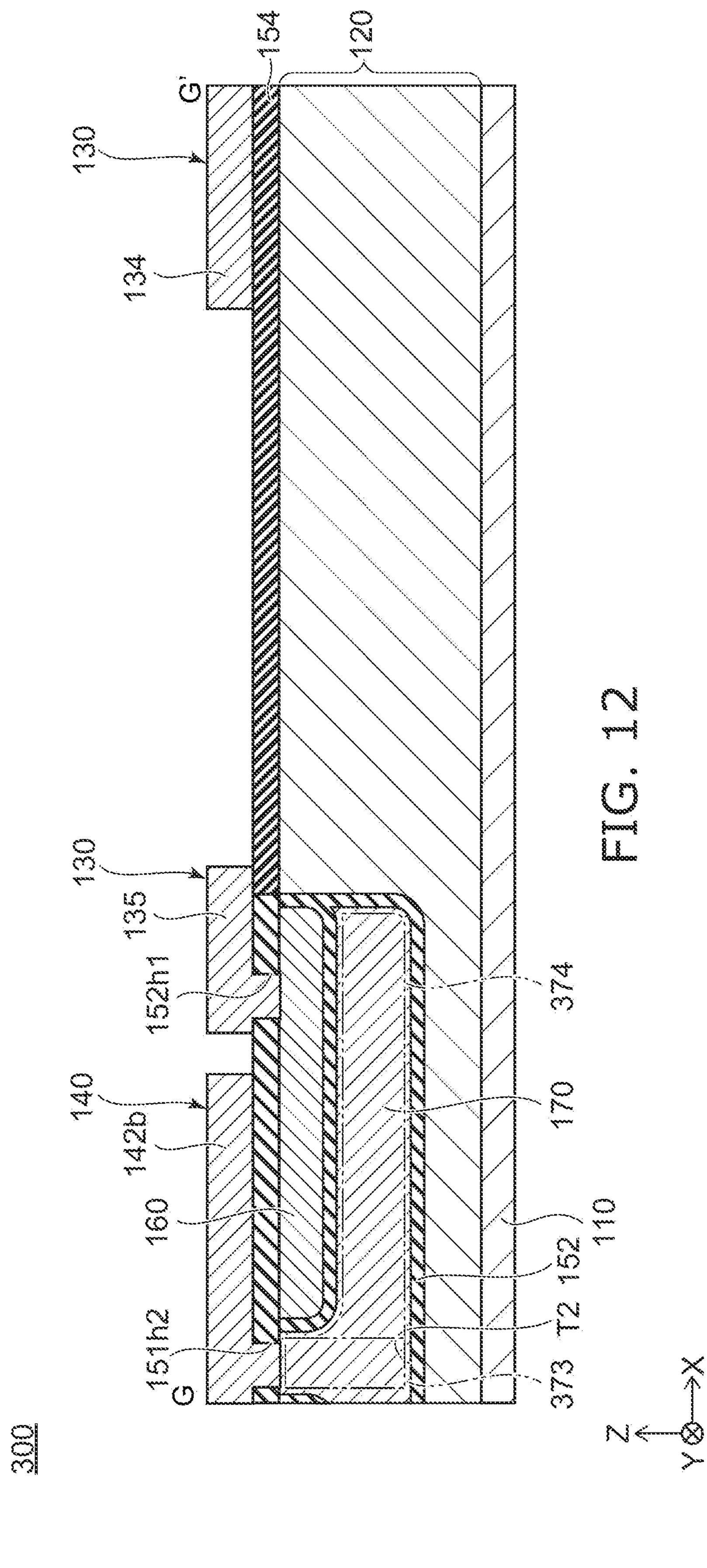
151h2
154
120
372
170
110
151
T1
371
Z
Y
X

FIG. 11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130
134
G'
154
120
130
135
152h1
140
142b
160
151h2
G
374
170
110
152
T2
373
Z
Y
X

FIG. 12

SEMICONDUCTOR DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. application Ser. No. 17/669,309, filed on Feb. 10, 2022, the entire contents of which are incorporated herein by reference, and which is based upon and claims the benefit of priority from Japanese Patent Application No. 2021-150214, filed on Sep. 15, 2021; the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiments relate to a semiconductor device.

BACKGROUND

[0003] In a conventionally known structure of a semiconductor device such as a MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor) or the like in which a gate electrode and a FP (Field Plate) electrode are located inside a semiconductor part, an external gate electrode that is electrically connected to the gate electrode inside the semiconductor part and a source electrode that is electrically connected to the FP electrode are located at the surface of the semiconductor part.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. **1** is a top view showing a semiconductor device according to a first embodiment;

[0005] FIG. **2** is an enlarged top view of a portion of FIG. **1** surrounded with broken line A;

[0006] FIG. **3** is a cross-sectional view along line B-B' of FIG. **2**;

[0007] FIG. **4** is a cross-sectional view along line C-C' of FIG. **2**;

[0008] FIG. **5** is a cross-sectional view along line D-D' of FIG. **2**;

[0009] FIG. **6** is a schematic view showing a positional relationship of components of the semiconductor device according to the first embodiment when viewed from above;

[0010] FIG. **7** is an enlarged schematic view showing the positional relationship of components of a semiconductor device according to a reference example when viewed from above;

[0011] FIG. **8** is a top view showing a semiconductor device according to a second embodiment;

[0012] FIG. **9** is a schematic view showing a positional relationship of components when viewed from above in the portion surrounded with broken line E of FIG. **8**;

[0013] FIG. **10** is a schematic view showing a positional relationship of components of a semiconductor device according to a third embodiment when viewed from above;

[0014] FIG. **11** is a cross-sectional view along line F-F' of FIG. **10**; and

[0015] FIG. **12** is a cross-sectional view along line G-G' of FIG.

DETAILED DESCRIPTION

[0016] In general, according to one embodiment, a semiconductor device includes: a first electrode; a semiconductor part located on the first electrode; a second electrode located in the semiconductor part, the second electrode extending in a first direction when viewed from above; a third electrode located in the semiconductor part, the third electrode extending in the first direction, the third electrode including a first portion, a second portion, and a first middle portion positioned below the second electrode between the first portion and the second portion, the second electrode being located between the first portion and the second portion in the first direction; a fourth electrode located above the semiconductor part, the fourth electrode including a pad portion separated from the second electrode and the second portion in a second direction crossing the first direction when viewed from above, and a protrusion protruding from the pad portion and covering the second electrode, the protrusion being connected to the second electrode; a fifth electrode located above the semiconductor part and separated from the fourth electrode, the fifth electrode including a first covering portion covering a first contact portion of the semiconductor part adjacent to the second portion when viewed from above, the first covering portion being connected to the first contact portion, and a second covering portion covering the first portion, the second covering portion being connected to the first portion; and a first insulating film located between the semiconductor part and the second electrode, between the semiconductor part and the third electrode, and between the second electrode and the third electrode.

[0017] In general, according to one embodiment, a semiconductor device includes: a first electrode; a semiconductor part located on the first electrode; a second electrode located in the semiconductor part, the second electrode extending in a first direction when viewed from above; a third electrode, the second electrode including a first end portion in an opposite direction of the first direction, the third electrode including a portion next to the first end portion; a fourth electrode located in the semiconductor part, the fourth electrode being next to the first end portion of the second electrode in a second direction crossing the first direction when viewed from above, the fourth electrode extending in the first direction, a length in the first direction of the fourth electrode being less than a length in the first direction of the second electrode; a fifth electrode, the fourth electrode including a second end portion in the opposite direction of the first direction, the fifth electrode including a portion next to the second end portion in the first direction; a sixth electrode located above the semiconductor part, the sixth electrode being connected to the fourth electrode, the sixth electrode including a pad portion and a protrusion, the pad portion being separated from the second and fourth electrodes in the second direction, the protrusion protruding from the pad portion, the fourth electrode including a third end portion in the first direction, the protrusion covering the third end portion, the protrusion being connected to the third end portion; and a seventh electrode located above the semiconductor part, the seventh electrode being separated from the pad portion in an opposite direction of the second direction, the seventh electrode including a first covering portion and a second covering portion, a distance from the pad portion to the first covering portion in the second direction being greater than a distance between the fourth electrode and the pad portion in the second direction, the second covering portion covering the portion of the third electrode, the portion of the fifth electrode, and a first contact portion of the semiconductor part, the first contact portion being positioned between the second electrode and the fourth electrode, the second covering portion being connected to the portion of the third electrode, the portion of the fifth electrode, and the first contact portion; a first insulating film located between the semiconductor part and the second electrode, between the semiconductor part and the third electrode, and between the second electrode and the third electrode; and a second insulating film located between the semiconductor part and the fourth electrode, between the semiconductor part and the fifth electrode, and between the fourth electrode and the fifth electrode.

[0018] In general, according to one embodiment, a semiconductor device includes: a first electrode; a semiconductor part located on the first electrode; a second electrode located in the semiconductor part, the second electrode extending in a first direction when viewed from above; a third electrode located in the semiconductor part, the third electrode extending in the first direction, the third electrode including a first portion, a second portion, and a first middle portion, the second electrode including a first end portion in an opposite direction of the first direction, the first portion being next to the first end portion in the first direction, the second electrode including a second end portion in the first direction, the second portion being next to the second end portion in the first direction, the first middle portion being positioned below the second electrode between the first portion and the second portion; a fourth electrode located in the semiconductor part, the fourth electrode being next to the second electrode in a second direction crossing the first direction when viewed from above, the fourth electrode extending in the first direction; a fifth electrode located in the semiconductor part, the fourth electrode including a third end portion in the opposite direction of the first direction, the fifth electrode including a third portion that is next to the third end portion in the first direction; a sixth electrode located above the semiconductor part, the sixth electrode including a pad portion and a protrusion, the pad portion being separated in the second direction from the second and fourth electrodes, the protrusion protruding from the pad portion, the fourth electrode including a fourth end portion in the first direction, the protrusion covering the second end portion of the second electrode and the fourth end portion of the fourth electrode, the protrusion being connected to the second and fourth end portions, the protrusion not covering at least a portion of the second portion when viewed from above; a seventh electrode located above the semiconductor part and separated from the pad portion in an opposite direction of the second direction, the seventh electrode including a first covering portion and a second covering portion, a distance from the pad portion to the first covering portion in the second direction being greater than a distance between the fourth electrode and the pad portion in the second direction, the second covering portion covering the first portion, the third portion, and a first contact portion of the semiconductor part positioned between the second electrode and the fourth electrode, the second covering portion being connected to the first portion, the third portion, and the first contact portion; a first insulating film located between the semiconductor part and the second electrode, between the semiconductor part and the third electrode, and between the second electrode and the third electrode; and a second insulating film located between the semiconductor part and the fourth electrode, between the semiconductor part and the fifth electrode, and between the fourth electrode and the fifth electrode.

[0019] Exemplary embodiments will now be described with reference to the drawings. The drawings are schematic or conceptual; and the relationships between the thickness and width of portions, the proportional coefficients of sizes among portions, etc., are not necessarily the same as the actual values Furthermore, the dimensions and proportional thereof. coefficients may be illustrated differently among drawings, even for identical portions. In the specification of the application and the drawings, components similar to those described in regard to a drawing thereinabove are marked with like reference numerals, and a detailed description is omitted as appropriate.

[0020] For easier understanding of the following description, the arrangements and configurations of the portions are described using an XYZ orthogonal coordinate system. An X-axis, a Y-axis, and a Z-axis are orthogonal to each other. The direction in which the X-axis extends is taken as an "X-direction"; the direction in which the Y-axis extends is taken as a "Y-direction"; and the direction in which the Z-axis extends is taken as a "Z-direction". The direction of the arrow in the X-direction is taken as the "+X direction"; and the opposite direction is taken as the "−X direction". The direction of the arrow in the Y-direction is taken as the "+Y direction"; and the opposite direction is taken as the "−Y direction". Although the direction of the arrow in the Z-direction is taken as up and the opposite direction is taken as down, these directions are independent of the direction of gravity.

[0021] Hereinbelow, the notations of the + and − for each conductivity type indicate relative levels of the impurity concentrations of each conductivity type. Specifically, a notation marked with "+" indicates a higher impurity concentration than a notation not marked with either "+" or "−". A notation marked with E indicates a lower impurity concentration than a notation not marked with either "+" or "−". Here, when both an impurity that forms donors and an impurity that forms acceptors are included in each region, the "impurity concentration" means the net impurity concentration after the impurities cancel.

First Embodiment

[0022] First, a first embodiment will be described.

[0023] FIG. **1** is a top view showing a semiconductor device according to the embodiment.

[0024] FIG. **2** is an enlarged top view of a portion of FIG. **1** surrounded with broken line A.

[0025] FIG. **3** is a cross-sectional view along line B-B' of FIG. **2**.

[0026] FIG. **4** is a cross-sectional view along line C-C' of FIG. **2**.

[0027] FIG. **5** is a cross-sectional view along line D-D' of FIG. **2**.

[0028] FIG. **6** is a schematic view showing the positional relationship of the components of the semiconductor device according to the embodiment when viewed from above.

[0029] In FIG. **6**, components that are actually not coplanar are shown as being coplanar to illustrate the positional relationship of the components of the semiconductor device **100** according to the embodiment when viewed from above. Also, for easier understanding of the areas in which a FP electrode **170**, a p-type base region **122***a*, a p^+-type contact region **122***b*, and an n-type source layer **123** that are described below are located in FIG. **6**, the areas in which the FP electrode **170**, the p-type base region **122***a*, the p^+-type contact region **122***b*, and the n-type source layer **123** are located are shown by diagonal hatching.

[0030] The semiconductor device **100** according to the embodiment is, for example, a MOSFET. However, it is sufficient for the semiconductor device to include a gate electrode and a FP electrode; and the semiconductor device may be, for example, an IGBT (Insulated Gate Bipolar Transistor).

[0031] As shown in FIG. **1**, the shape of the semiconductor device **100** when viewed from above is substantially a rectangle of which each side is substantially parallel to the X-direction or the Y-direction. However, the shape of the semiconductor device is not limited to that described above. Generally speaking, as shown in FIGS. **2** and **3**, the semiconductor device **100** includes a drain electrode **110**, a semiconductor part **120**, an external gate electrode **130**, and a source electrode **140**. The semiconductor device **100** further includes an insulating film **151**, multiple insulating films **152**, multiple insulating films **153**, and an insulating film **154**. The semiconductor device further includes multiple internal gate electrodes **160** and multiple FP electrodes **170**. The components of the semiconductor device **100** will now be described. Hereinbelow, the configuration of a pad portion **131** of the external gate electrode **130** shown in FIG. **1** and the vicinity of the pad portion **131** in the semiconductor device **100** is mainly described, and a description of the other structures is omitted as appropriate.

[0032] The drain electrode **110** is located in substantially the entire region of the lower surface of the semiconductor device **100**. For example, as shown in FIG. **3**, the upper surface and the lower surface of the drain electrode **110** are flat surfaces that are substantially parallel to the XY plane. The drain electrode **110** is made of a conductive material such as a metal material, etc. The semiconductor part **120** is located on the drain electrode **110**.

[0033] The semiconductor part **120** includes, for example, an n-type semiconductor layer **121**, a p-type semiconductor layer **122** that is located in the upper layer portion of the n-type semiconductor layer **121**, and the n^+-type source layer **123** that is located in the upper layer portion of the p-type semiconductor layer **122**. However, the configuration of the semiconductor part can be modified as appropriate according to the type of element.

[0034] The n-type semiconductor layer **121** includes an n-type drift region. The n-type drift region is located on substantially the entire region of the upper surface of the drain electrode **110**. However, the n-type semiconductor layer may further include an n^+-type drain region that is located between the n-type drift region and the drain electrode and includes a higher impurity concentration than the n^--type drift region.

[0035] The p-type semiconductor layer **122** includes the p-type base region **122***a*, and the p^+-type contact region **122***b* that is located in the p-type base region **122***a*. As shown in FIG. **5**, the p-type base region **122***a* is partially provided in the upper layer portion of the n-type semiconductor layer **121**. The p^+-type contact region **122***b* is partially provided in the p-type base region **122***a* and has an ohmic contact with the source electrode **140** that is described below. The details of the arrangement of the p-type base region **122***a*, the p^+-type contact region **122***b*, and the n^+-type source layer **123** are described below.

[0036] As shown in FIG. **1**, the external gate electrode **130** and the source electrode **140** that is separated from the external gate electrode **130** are located above the semiconductor part **120**.

[0037] The external gate electrode **130** has a so-called finger configuration. The external gate electrode **130** includes, for example, the pad portion **131**, a first extension portion **132**, multiple second extension portions **133**, a third extension portion **134**, and a protrusion **135**.

[0038] The pad portion **131** is a portion to which a wiring member (not illustrated) is connected to electrically connect the external gate electrode **130** to an external circuit of the semiconductor device **100**. According to the embodiment, the pad portion **131** is located at the corner among the four corners of the semiconductor part **120** that is positioned furthest toward the +X side and the +Y side. According to the embodiment, the pad portion **131** is substantially rectangular when viewed from above.

[0039] The first extension portion **132** extends in the −X direction from the corner of the pad portion **131** that is positioned furthest toward the −X side and the +Y side. The first extension portion **132** is positioned on the +Y direction end portion of the semiconductor part **120**. The second extension portions **133** extend in the −Y direction from the first extension portion **132**. The multiple second extension portions **133** are arranged in the X-direction. The third extension portion **134** extends in the −Y direction from the corner of the pad portion **131** that is positioned furthest toward the +X side and the −Y side. The −Y direction end portions of each of the second and third extension portions **133** and **134** extend to the vicinity of the −Y direction end portion of the semiconductor part **120**.

[0040] The protrusion **135** protrudes in the −Y direction from the −X direction end portion of the pad portion **131**, and specifically, the corner positioned furthest toward the −X side and the −Y side. The shape of the protrusion **135** when viewed in top-view is, for example, substantially rectangular. The length of the protrusion **135** in the Y-direction is less than the lengths of the second and third extension portions **133** and **134** in the Y-direction.

[0041] The external gate electrode **130** is made of a conductive material such as a metal material, etc. The external gate electrode **130** may include a stacked body of two or more metal layers.

[0042] As shown in FIG. **1**, the source electrode **140** also has a so-called finger configuration. The source electrode **140** includes, for example, multiple first extension portions **141**, a second extension portion **142**, and multiple third extension portions **143**.

[0043] The first extension portions **141** and the second extension portion **142** extend in the +Y direction from the −Y direction end portion of the semiconductor part **120**. Each first extension portion **141** is positioned between two second extension portions **133** of the external gate electrode **130** that are next to each other. The second extension portion **142** is positioned between the second extension portion **133** and the third extension portion **134** that are next to each other. Each third extension portion **143** is positioned between the −Y direction end portions of two second extension portions **142** that are next to each other or between the −Y direction end portion of the second extension portion **142** and the −Y direction end portion of the third extension portion **143** that are next to each other; and the third extension portions **143** link these portions to each other.

[0044] The second extension portion **142** includes a first covering portion **142***a*, a second covering portion **142***b*, and a third covering portion **142***c*. The third covering portion **142***c* extends in the +Y direction from the −Y direction end portion of the semiconductor part **120**. The first covering portion **142***a* and the second covering portion **142***b* protrude in the +Y direction from the third covering portion **142***c*. The first covering portion **142***a* is positioned at the −Y side of the pad portion **131** and the +X side of the protrusion **135**. The second covering portion **142***b* is positioned at the −X side of the pad portion **131** and the protrusion **135**. The protrusion **135** is positioned between the first covering portion **142***a* and the second covering portion **142***b* in the X-direction. The shapes of the first covering portion **142***a*, the second covering portion **142***b*, and the third covering portion **142***c* when viewed from above are, for example, rectangular. A recess **144** is formed from the first covering portion **142***a*, the second covering portion **142***b*, and the third covering portion **142***c*. The protrusion **135** is located in the recess **144**. Here, "the protrusion is located in the recess" means that at least a portion of the protrusion is positioned in the recess.

[0045] The source electrode **140** is made of a conductive material such as a metal material, etc. The source electrode **140** may include a stacked body of two or more metal layers.

[0046] As shown in FIGS. **2** and **3**, a trench T**1**, two trenches T**2**, two trenches T**3**, multiple trenches T**4**, and multiple trenches T**5** are provided in the semiconductor part **120**. The trenches T**1** to T**5** extend in the X-direction and are so-called straight-type trenches without branches. The trenches T**1** to T**5** extend downward from the upper surface of the semiconductor part **120**. The number of the trenches T**2** and the number of the trenches T**3** are not limited to two and may be one, three, or more.

[0047] As shown in FIG. **6**, a portion of the trench T**1** is covered with the second covering portion **142***b*, the protrusion **135**, and the first covering portion **142***a* when viewed from above. The +X direction end portion of the trench T**1** is covered with the third extension portion **134** when viewed from above.

[0048] The insulating film **151** is located in the trench T**1**. The insulating film **151** is made of an insulating material such as silicon oxide, silicon nitride, etc. The multiple internal gate electrodes **160** and the FP electrode **170** are located in the insulating film **151**. The internal gate electrode **160** that is positioned furthest toward the +X side among the multiple internal gate electrodes **160** in the trench T**1** will now be described.

[0049] As shown in FIGS. **4** and **6**, the internal gate electrode **160** that is in the trench T**1** extends in the X-direction. When viewed from above, the −X direction end portion of the internal gate electrode **160** is covered with the second covering portion **142***b*; and the +X direction end portion of the internal gate electrode **160** is covered with the protrusion **135**. An opening **151***h***1** is provided in the portion of the insulating film **151** positioned between the internal gate electrode **160** and the protrusion **135**. The protrusion **135** is connected to the internal gate electrode **160** via the opening **151***h***1**. Thereby, the external gate electrode **130** is connected to the internal gate electrode **160** via a current path that extends in the Z-direction.

[0050] As shown in FIGS. **4** and **6**, the FP electrode **170** that is in the trench T**1** extends in the X-direction. The FP electrode **170** includes a first portion **171**, a second portion **172**, and a first middle portion **173** positioned between the first portion **171** and the second portion **172**, such that the internal gate electrode **160** is positioned between the first portion **171** and the second portion **172** in the X-direction, and the first middle portion **173** is positioned below the internal gate electrode **160**. The first portion **171** is positioned at the −X side of the internal gate electrode **160**. In other words, the first portion **171** is next to the −X direction end portion of the internal gate electrode **160** in the X-direction. The second portion **172** is positioned at the +X side of the internal gate electrode **160**, and according to the embodiment, includes the +X direction end portion of the FP electrode **170**. In other words, the second portion **172** is next to the +X direction end portion of the internal gate electrode **160** in the X-direction. The insulating film **151** is located between the FP electrode **170** and the internal gate electrode **160**.

[0051] Thus, at the vicinity of the pad portion **131**, the FP electrode **170** separates the internal gate electrode **160** from the other internal gate electrode **160** (not illustrated) that is positioned further toward the −X side than the first portion **171** of the FP electrode **170**. According to the embodiment, the internal gate electrode **160** thus separated is connected to the protrusion **135**. Therefore, the internal gate electrode **160** thus separated can be electrically connected to the external gate electrode **130**.

[0052] The first portion **171** is covered with the second covering portion **142***b*. An opening **151***h***2** is provided in the portion of the insulating film **151** positioned between the first portion **171** and the second covering portion **142***b*. The second covering portion **142***b* is connected to the first portion **171** via the opening **151***h***2**.

[0053] The −X direction end portion of the second portion **172** is covered with the protrusion **135**; and the remaining portion of the second portion **172** is not covered with the protrusion **135** when viewed from above. The +X direction end portion of the second portion **172** is covered with the third extension portion **134**. The portion of the second portion **172** positioned between the two X-direction end portions is covered with the first covering portion **142***a*. An opening **151***h***3** is provided in the portion of the insulating film **151** positioned between the second portion **172** and the first covering portion **142***a*. The first covering portion **142***a* is connected to the second portion **172** via the opening **151***h***3**. Thereby, the potential of the second portion **172** extending from the protrusion **135** to the third extension portion **134** when viewed from above easily has the same potential as the source electrode **140**. However, because the FP electrode **170** is electrically connected to the source electrode **140** via the first portion **171**, the first covering portion **142***a* may not be connected to the second portion **172** via the opening **151***h***3** of the insulating film **151**.

[0054] The two trenches T**2** are positioned at the +Y side of the trench T**1** and are arranged in the Y-direction. When viewed from above, a portion of each trench T**2** is covered with the second covering portion **142***b*; and the +X direction end portion of each trench T**2** is covered with the protrusion **135**. The trenches T**2** are not covered with the first covering portion **142***a*.

[0055] The insulating film **152** is located in each trench T**2**. The insulating films **152** are made of an insulating material such as silicon oxide, silicon nitride, etc. The multiple internal gate electrodes **160** and the FP electrode **170** are located in each insulating film **152**. The internal gate electrode **160** that is positioned furthest toward the +X side among the multiple internal gate electrodes **160** in each trench T**2** will now be described.

[0056] The internal gate electrode **160** that is in each trench T**2** extends in the X-direction. When viewed from above, the −X direction end portion of this internal gate electrode **160** is covered with the second covering portion **142***b*; and the +X direction end portion of this internal gate electrode **160** is covered with the protrusion **135**. An opening **152***h***1** is provided in the portion of the insulating film **152** positioned between this internal gate electrode **160** and the protrusion **135**. The protrusion **135** is connected to this internal gate electrode **160** via the opening **152***h***1**.

[0057] The FP electrode **170** that is in each trench T**2** extends in the X-direction. As shown in FIGS. **3** and **6**, the FP electrode **170** includes a third portion **174**, a fourth portion **175**, and a second middle portion **176** positioned between the third portion **174** and the fourth portion **175**, such that the internal gate electrode **160** is positioned between the third portion **174** and the fourth portion **175** in the X-direction, and the second middle portion **176** is positioned below the internal gate electrode **160**. The third portion **174** is positioned at the −X side of the internal gate electrode **160**. In other words, the third portion **174** is next to the −X direction end portion of the internal gate electrode **160** in the X-direction. The third portion **174** is next to the first portion **171** in the Y-direction. The fourth portion **175** is positioned at the +X side of the internal gate electrode **160**, and according to the embodiment, corresponds to the +X direction end portion of the FP electrode **170**. In other words, the fourth portion **175** is next to the +X direction end portion of the internal gate electrode **160** in the X-direction. The insulating film **152** is located between the internal gate electrode **160** and the FP electrode **170**.

[0058] Thus, at the vicinity of the pad portion **131**, the FP electrode **170** separates the internal gate electrode **160** from the other internal gate electrode **160** (not illustrated) that is positioned further toward the −X side than the third portion **174** of the FP electrode **170**. According to the embodiment, the internal gate electrode **160** thus separated is connected to the protrusion **135**. Therefore, the internal gate electrode **160** thus separated can be electrically connected to the external gate electrode **130**.

[0059] The third portion **174** is covered with the second covering portion **142***b*. An opening **152***h***2** is provided in the portion of the insulating film **152** positioned between the third portion **174** and the second covering portion **142***b*. The second covering portion **142***b* is connected to the third portion **174** via the opening **152***h***2**. The fourth portion **175** is covered with the protrusion **135**. The length in the X-direction of the fourth portion **175** is less than the length in the X-direction of the second portion **172**.

[0060] Thus, in the +Y direction, the pad portion **131** is separated from the +X direction end portion of the internal gate electrode **160** in each trench T**2**. A distance L between the first covering portion **142***a* and the pad portion **131** in the Y-direction is greater than distances L**1** and L**2** between the pad portion **131** and the trenches T**2** in the Y-direction. In other words, the positions of the trenches T**2** in the Y-direction are between the position of the pad portion **131** in the Y-direction and the position of the first covering portion **142***a* in the Y-direction. The protrusion **135** protrudes from the pad portion **131**, covers the +X direction end portion of the internal gate electrode **160** in each trench T**2**, and is connected to the +X direction end portion of the internal gate electrode **160** in each trench T**2**.

[0061] The two trenches T**3** are positioned at the −Y side of the trench T**1** in the Y-direction. When viewed from above, a portion of each trench T**3** is covered with the first and second covering portions **142***a* and **142***b* and not covered with the protrusion **135**. The +X direction end portion of each trench T**3** is covered with the third extension portion **134** when viewed from above.

[0062] The insulating film **153** is located in each trench T**3**. The insulating films **153** are made of an insulating material such as silicon oxide, silicon nitride, etc. The multiple internal gate electrodes **160** and the FP electrode **170** are located in each insulating film **153**. The internal gate electrode **160** that is positioned furthest toward the +X side among the multiple internal gate electrodes **160** in each trench T**3** will now be described.

[0063] The internal gate electrode **160** that is in each trench T**3** extends in the X-direction. When viewed from above, the −X direction end portion of this internal gate electrode **160** is covered with the second covering portion **142***b*; and the +X direction end portion of this internal gate electrode **160** is covered with the third extension portion **134**. The first covering portion **142***a* covers between the two X-direction end portions of this internal gate electrode **160**. An opening **153***h***1** is provided in the portion of the insulating film **153** positioned between this internal gate electrode **160** and the third extension portion **134**. The third extension portion **134** is connected to this internal gate electrode **160** via the opening **153***h***1**.

[0064] The FP electrode **170** that is in each trench T**3** extends in the X-direction. As shown in FIGS. **3** and **6**, the FP electrode **170** includes a fifth portion **177**, a sixth portion **178**, and a third middle portion **179** between the fifth portion **177** and the sixth portion **178**, such that the internal gate electrode **160** is positioned between the fifth portion **177** and the sixth portion **178** in the X-direction, and the third middle portion **179** is positioned below the internal gate electrode **160**. The fifth portion **177** is positioned at the −X side of the internal gate electrode **160**. In other words, the fifth portion **177** is next to the −X direction end portion of the internal gate electrode **160** in the X-direction. The fifth portion **177** is next to the first portion **171** in the Y-direction. The sixth portion **178** is positioned at the +X side of the internal gate electrode **160**, and according to the embodiment, corresponds to the +X direction end portion of the FP electrode **170**. In other words, the sixth portion **178** is next to the +X direction end portion of the internal gate electrode **160** in the X-direction. The insulating film **153** is located between the internal gate electrode **160** and the FP electrode **170**.

[0065] Thus, at the vicinity of the pad portion **131**, the FP electrode **170** separates the internal gate electrode **160** from the other internal gate electrode **160** (not illustrated) that is positioned further toward the −X side than the fifth portion **177** of the FP electrode **170**. According to the embodiment, the internal gate electrode **160** thus separated is connected to the third extension portion **134**. Therefore, the internal gate electrode **160** thus separated can be electrically connected to the external gate electrode **130**.

[0066] The fifth portion **177** is covered with the second covering portion **142***b*. An opening **153***h***2** is provided in the portion of the insulating film **153** positioned between the fifth portion **177** and the second covering portion **142***b*. The second covering portion **142***b* is connected to the fifth portion **177** via the opening **153**h**2**. The sixth portion **178** is covered with the third extension portion **134**. The length in the X-direction of the sixth portion **178** is less than the length in the X-direction of the second portion **172**.

[0067] The multiple trenches T**4** are located at the +Y side of the multiple trenches T**2** and are arranged in the Y-direction. Other than the X-direction end portion being covered with the pad portion **131** instead of the protrusion **135** when viewed from above, the configuration and the internal configuration of each trench T**4** are substantially similar to those of the trench T**2**; and a description is therefore omitted.

[0068] The multiple trenches T**5** are located at the +Y side of the multiple trenches T**3** and are arranged in the Y-direction. Other than being covered with the third covering portion **142**c and not with the first and second covering portions **142**a and **142**b when viewed from above, the configuration and the internal configuration of each trench T**5** are substantially similar to those of the trench T**3**; and a description is therefore omitted.

[0069] As shown in FIG. **5**, the insulating film **154** is located between the external gate electrode **130** and the upper surface of the semiconductor part **120** and between the source electrode **140** and the upper surface of the semiconductor part **120**.

[0070] As shown in FIG. **6**, three openings **154**h**1**, multiple openings **154**h**2**, and multiple openings **154**h**3** are provided in the insulating film **154**.

[0071] Each opening **154**h**1** is positioned between the first covering portion **142**a of the source electrode **140** and a contact portion **120**a positioned between the trenches T**1**, T**3**, and T**5** that are next to each other in the semiconductor part **120**. The openings **154**h**1** extend in the X-direction and are covered with the first covering portion **142**a when viewed from above. As shown in FIG. **5**, the first covering portion **142**a is connected to the contact portions **120**a via the openings **154**h**1**. According to the embodiment, the contact portion **120**a is the surface of a trench T**6** that extends downward from the upper surface of the semiconductor part **120**. This is similar for the other contact portions **120**b and **120**c that are described below as well. However, a portion of the upper surface of the semiconductor part may be used as a contact portion without providing such a trench in the semiconductor part.

[0072] According to the embodiment, the p-type base region **122**a and the p^+-type contact region **122**b are included but the n-type source layer **123** is not included in the contact portion **120**a between the trench T**1** and the trench T**3** that are next to each other and in the contact portion **120**a between the trenches T**3** that are next to each other. However, the n-type source layer **123** also may be included in the contact portion **120**a. Also, according to the embodiment, the p-type base region **122**a, the p^+-type contact region **122**b, and the n-type source layer **123** are included in the contact portion **120**a between the trench T**3** and the trench T**5** that are next to each other.

[0073] As shown in FIG. **6**, each opening **154**h**2** is positioned between the second covering portion **142**b of the source electrode **140** and the contact portion **120**b positioned between the trenches T**1**, T**2**, and T**4** that are next to each other in the semiconductor part **120**. The openings **154**h**2** extend in the X-direction. As shown in FIG. **3**, the second covering portion **142**b is connected to the contact portions **120**b via the openings **154**h**2**. According to the embodiment, the p-type base region **122**a, the p^+-type contact region **122**b, and the n-type source layer **123** are included in each contact portion **120**b.

[0074] As shown in FIG. **6**, each opening **154**h**3** is positioned between the third covering portion **142**c and the contact portion **120**c positioned between the trenches T**5** that are next to each other in the semiconductor part **120**. The openings **154**h**3** extend in the X-direction. As shown in FIG. **3**, the third covering portion **142**c is connected to the contact portions **120**c via the openings **154**h**3**. According to the embodiment, the p-type base region **122**a, the p^+-type contact region **122**b, and the n-type source layer **123** are included in each contact portion **120**c.

[0075] Effects of the embodiment will now be described.

[0076] FIG. **7** is an enlarged schematic view showing the positional relationship of components of a semiconductor device according to a reference example when viewed from above.

[0077] In the semiconductor device **900** according to the reference example, the protrusion **135** that protrudes in the −Y direction from the pad portion **131** is not included in an external gate electrode **930**. Also, in the semiconductor device **900** according to the reference example, the recess **144** in which the protrusion **135** is located is not provided in a source electrode **940**.

[0078] It is necessary to separate the external gate electrode **930** and the source electrode **940** by the distance L of some length in the Y-direction. An example is described hereinbelow in which the distance L is the length between two trenches arranged in the Y-direction.

[0079] The vicinity of the pad portion **131** is the termination region of the semiconductor device **900**. Therefore, from the perspective of increasing the breakdown voltage, it is favorable for the FP electrode **170** to extend to the +X direction end portion of each of the trenches T**1** to T**6**, and for the internal gate electrode **160** not to be provided in the end portion of each of the trenches T**1** to T**6**. Thus, when the FP electrode **170** extends to the +X direction end portion of each of the trenches T**1** to T**6**, even if the internal gate electrode **160** is located at the vicinity of the pad portion **131** in each of the trenches T**1** to T**6**, the FP electrode **170** separates this internal gate electrode **160** from the internal gate electrode **160** at the center of the semiconductor part **120**. Accordingly, in the semiconductor device **900** according to the reference example, it is necessary to connect such an internal gate electrode **160** to the pad portion **131** or the third extension portion **134**.

[0080] However, in the semiconductor device **900** according to the reference example, even if the two trenches T**2** extend to be directly under the third extension portion **134**, the internal gate electrodes **160** are provided inside the two trenches T**2**, and the third extension portion **134** is connected to these internal gate electrodes **160**, the source electrode **940** cannot be connected to the portion of the semiconductor part **120** positioned between the pad portion **131** and the source electrode **940** in the Y-direction. Therefore, there is a possibility that the gate capacity and the avalanche resistance may decrease. Accordingly, in the semiconductor device **900** according to the reference example, it is favorable to provide the FP electrodes **170** in the +X direction end portions of the two trenches T**2** and cover the FP electrodes **170** with the source electrode **140** without the two trenches

T**2** extending to the third extension portion **134** so that the gate capacity and/or the avalanche resistance does not decrease.

[0081] In such a case, the trench T**1** and the portion of the trench T**4** that is positioned furthest toward the −Y side and is not positioned between two other trenches are end point regions; therefore, it is favorable to provide the FP electrode **170** from the perspective of increasing the breakdown voltage. Accordingly, in the semiconductor device **900** according to the reference example, the region between the trench T**1** and the trench T**4** positioned furthest toward the −Y side is ineffective as an element such as a MOSFET, etc.

[0082] Conversely, in the semiconductor device **100** according to the embodiment, the external gate electrode **130** includes the protrusion **135** that protrudes from the pad portion **131**, covers the internal gate electrode **160** in the trench T**1**, and is connected to the internal gate electrode **160**. The source electrode **140** includes the first covering portion **142***a* that covers the contact portion **120***a* of the semiconductor part **120** adjacent to the second portion **172** of the FP electrode in the trench T**1** when viewed from above and is connected to the contact portion **120***a*, and includes the second covering portion **142***b* that covers the first portion **171** and is connected to the contact portion **120***a*. Therefore, the internal gate electrode **160** can be located in the trench T**1** and connected to the external gate electrode **130** at the vicinity of the pad portion **131** while maintaining the state in which the FP electrode in the trench T**1** and the contact portion **120***a* of the semiconductor part **120** are connected to the source electrode **140**. Therefore, the effective element area can be increased.

[0083] The first covering portion **142***a* also covers the second portion **172** of the FP electrode **170** in the trench T**1** and is connected to the second portion **172**. Therefore, the potential of the second portion **172** can easily be the same potential as the potential of the source electrode **140**.

[0084] The internal gate electrode **160** that is in each trench T**2** positioned in the Y-direction between the position of the pad portion **131** in the Y-direction and the position of the first covering portion **142***a* in the Y-direction cannot be directly connected to the pad portion **131**. Conversely, the protrusion **135** covers the internal gate electrode **160** in the trench T**2** when viewed from above and is connected to the internal gate electrode **160**. Therefore, the effective element area can be increased.

[0085] The second covering portion **142***b* covers the contact portion **120***b* of the semiconductor part **120** positioned between the internal gate electrode **160** in the trench T**1** and the internal gate electrode **160** in the trench T**2** when viewed from above, and is connected to the contact portion **120***b*. Therefore, the effective element area can be increased.

[0086] The n-type source layer **123** is included in the contact portion **120***b*. Therefore, the surface area in which the n-type source layer **123** is located can be increased.

[0087] The n-type source layer **123** also is located under the first covering portion **142***a*; and the first covering portion **142***a* is connected to the n-type source layer **123**. Therefore, the surface area in which the n-type source layer **123** is located can be increased.

Second Embodiment

[0088] A second embodiment will now be described.

[0089] FIG. **8** is a top view showing a semiconductor device according to the embodiment.

[0090] FIG. **9** is a schematic view showing the positional relationship of components when viewed from above in the portion surrounded with broken line E of FIG. **8**.

[0091] The semiconductor device **200** according to the embodiment differs from the semiconductor device **100** according to the first embodiment in that a pad portion **231** is positioned at substantially the X-direction center of the semiconductor part **120**.

[0092] As a general rule in the following description, only the differences with the first embodiment are described. Other than the items described below, the embodiment is similar to the first embodiment.

[0093] An external gate electrode **230** includes the pad portion **231**, a first extension portion **232**, a second extension portion **233**, a third extension portion **234**, a fourth extension portion **235**, a fifth extension portion **236**, and two protrusions **237**.

[0094] The pad portion **231** is positioned at substantially the X-direction center at the +Y direction end portion of the semiconductor part **120**. The shape of the pad portion **231** when viewed from above is substantially rectangular. Thus, the position of the pad portion is not limited to a corner of the semiconductor part.

[0095] The first extension portion **232** extends in the −X direction from the corner of the pad portion **231** positioned furthest toward the −X side and the +Y side. The second extension portion **233** extends in the +X direction from the corner of the pad portion **231** positioned furthest toward the +X side and the +Y side. The third extension portion **234** extends in the −Y direction from the −X direction end portion of the first extension portion **232**. The fourth extension portion **235** extends in the −Y direction from substantially the X-direction center at the +Y direction end portion of the pad portion **231**. The fifth extension portion **236** extends in the −Y direction from the +X direction end portion of the second extension portion **233**.

[0096] One protrusion **237** protrudes in the −Y direction from the −X direction end portion of the pad portion **231**, and specifically, the corner positioned furthest toward the −X side and the −Y side. Another protrusion **237** protrudes in the −Y direction from the +X direction end portion of the pad portion **231**, and specifically, the end portion positioned furthest toward the +X side and the −Y side.

[0097] A source electrode **240** includes a first extension portion **241**, a second extension portion **242**, and a third extension portion **243**.

[0098] The first extension portion **241** extends in the +Y direction from the −Y direction end portion of the semiconductor part **120** and is positioned between the third extension portion **234** and the fourth extension portion **235** of the external gate electrode **230**. The second extension portion **242** extends in the +Y direction from the −Y direction end portion of the semiconductor part **120** and is positioned between the fourth extension portion **235** and the fifth extension portion **236** of the external gate electrode **230**. The third extension portion **243** is positioned between the −Y direction end portion of the first extension portion **241** and the −Y direction end portion of the second extension portion **242** and is linked to these end portions.

[0099] The first extension portion **241** includes a first covering portion **241***a*, a second covering portion **241***b*, and a third covering portion **241***c*. The third covering portion **241***c* extends in the +Y direction from the −Y direction end portion of the semiconductor part **120**. The first covering portion **241***a* and the second covering portion **241***b* protrude in the +Y direction from the third covering portion **241***c*. The first covering portion **241***a* is positioned at the −Y side of the pad portion **231** and the +X side of the protrusion **237**. The second covering portion **241***b* is positioned at the −X side of the pad portion **231** and the protrusion **237**. One protrusion **237** is positioned between the first covering portion **241***a* and the second covering portion **241***b* in the X-direction. A recess **244** is formed from the first covering portion **241***a*, the second covering portion **241***b*, and the third covering portion **241***c*. The protrusion **237** is located in the recess **244**.

[0100] The shape of the second extension portion **242** is substantially symmetric with the first extension portion **241** with respect to a central axis C of the semiconductor device **200** extending in the Y-direction; and a description is therefore omitted.

[0101] The structure of the vicinity of the pad portion **231** at the −X side of the central axis C will now be described. The structure of the vicinity of the pad portion **231** at the +X side of the central axis C is symmetric with the structure of the vicinity of the pad portion **231** at the −X side of the central axis C when referenced to the central axis C; and a description is therefore omitted.

[0102] A trench T**21**, two trenches T**22**, two trenches T**23**, multiple trenches T**24**, and multiple trenches T**25** are provided in the semiconductor part **120**. Other than a middle portion instead of the +X direction end portion of the trench T**21** being covered with the fourth extension portion **235**, the structure and the internal structure of the trench T**21** are similar to those of the trench T**1** according to the first embodiment; and a description is therefore omitted. The structure and the internal structure of the trench T**22** are similar to those of the trench T**2** according to the first embodiment; and a description is therefore omitted. Other than a middle portion instead of the +X direction end portion of the trench T**23** being covered with the fourth extension portion **235**, and the FP electrode **170** being located below the internal gate electrode **160** in the portion of the trench T**23** positioned directly under the fourth extension portion **235**, the structure and the internal structure of the trench T**23** are similar to those of the trench T**3** according to the first embodiment; and a description is therefore omitted. The structure and the internal structure of the trench T**24** are similar to those of the trench T**4** according to the first embodiment; and a description is therefore omitted. Other than a middle portion instead of the +X direction end portion of the trench T**25** being covered with the fourth extension portion **235**, and the FP electrode **170** being located below the internal gate electrode **160** in the portion of the trench T**25** positioned directly under the fourth extension portion **235**, the structure and the internal structure of the trench T**25** are similar to those of the trench T**5** according to the first embodiment; and a description is therefore omitted.

[0103] The protrusion **237** covers the internal gate electrode **160** in the trench T**21** and the internal gate electrode **160** in the trench T**22** when viewed from above and is connected to these internal gate electrodes via the openings **151***h***1** and **152***h***1**. The fourth extension portion **235** covers the internal gate electrodes **160** in the trenches T**23** and T**25** and is connected to these internal gate electrodes via the openings **153***h***1**.

[0104] The first covering portion **241***a* covers the second portion **172** of the FP electrode **170** in the trench T**21** when viewed from above and is connected to the second portion via the opening **151***h***3**. The first covering portion **241***a* covers the contact portions **120***a* between the trenches T**21** and T**23** that are next to each other in the semiconductor part **120**, and is connected to these contact portions via the openings **154***h***1**.

[0105] When viewed from above, the second covering portion **241***b* covers the first portion **171** of the FP electrode **170** in the trench T**21**, the third portions **174** of the FP electrodes **170** in the trenches T**22** and T**24**, and the FP electrodes **170** in the trenches T**23** and T**25**, and is connected to these FP electrodes via the openings **151***h***2**, **152***h***2**, and **153***h***2**. When viewed from above, the second covering portion **241***b* covers the contact portions **120***b* between the trenches T**21**, T**22**, and T**24** that are next to each other in the semiconductor part **120**, and is connected to these contact portions via the openings **154***h***2**.

[0106] The third covering portion **241***c* covers the FP electrodes **170** in the trenches T**25** when viewed from above and is connected to these FP electrodes via the openings **153***h***2**. When viewed from above, the third covering portion **241***c* covers the contact portions **120***c* between the trenches T**23** and T**25** that are next to each other in the semiconductor part **120**, and is connected to these contact portions via the openings **154***h***3**.

[0107] Effects of the embodiment will now be described.

[0108] In the semiconductor device **200** according to the embodiment as well, the external gate electrode **230** includes the protrusion **237** that protrudes from the pad portion **231**, covers the internal gate electrode **160** in the trench T**21**, and is connected to the internal gate electrode **160**. The source electrode **240** includes the first covering portion **241***a* that covers the contact portion **120***a* of the semiconductor part **120** adjacent to the second portion **172** of the FP electrode in the trench T**21** when viewed from above and is connected to the second portion **172**, and includes the second covering portion **241***b* that covers the first portion **171** and is connected to the first portion **171**. Therefore, the internal gate electrode **160** can be connected to the external gate electrode **230** in the trench T**21** while maintaining the state in which the FP electrode **170** in the trench T**21** and the contact portion **120***a* of the semiconductor part **120** are connected to the source electrode **240**. Therefore, the effective element area can be increased.

Third Embodiment

[0109] A third embodiment will now be described.

[0110] FIG. **10** is a schematic view showing the positional relationship of components of the semiconductor device according to the embodiment when viewed from above.

[0111] FIG. **11** is a cross-sectional view along line F-F' of FIG. **10**.

[0112] FIG. **12** is a cross-sectional view along line G-G' of FIG. **10**.

[0113] The internal structures of the trenches T**1**, T**2**, T**3**, T**4**, and T**5** of the semiconductor device **300** according to the embodiment are different from those of the semiconductor device **100** according to the first embodiment.

[0114] According to the embodiment, the internal gate electrode **160** that is in the trench T**1** extends to a position that overlaps the third extension portion **134** when viewed from above. The +X direction end portion of this internal gate electrode **160** overlaps the third extension portion **134**. Accordingly, the contact portion **120***a* is adjacent to this internal gate electrode **160**. The length in the X-direction of the internal gate electrode **160** in the trench T**2** is less than the length in the X-direction of the internal gate electrode **160** in the trench T**1**.

[0115] According to the embodiment, the protrusion **135** overlaps a portion of this internal gate electrode **160** between the two X-direction end portions when viewed from above. Similarly to the first embodiment, the protrusion **135** is connected to the portion of this internal gate electrode **160** between the two X-direction end portions via an opening **151***h* in the insulating film **151** in the trench T**1**. However, this internal gate electrode **160** may be connected to the third extension portion **134** instead of the protrusion **135**. In such a case, it is sufficient to provide an opening in a portion of the insulating film **151** positioned between the third extension portion **134** and the +X direction end portion of this internal gate electrode **160**. Also, in such a case, the protrusion **135** may not cover the trench T**1**. In other words, the protrusion **135** may be separated from the trench T**1** in the +Y direction when viewed from above. This internal gate electrode **160** may be connected to both the protrusion **135** and the third extension portion **134**.

[0116] As shown in FIGS. **10** and **11**, the FP electrode **170** that is in the trench T**1** includes a portion **371** that is next to the −X direction end portion of the internal gate electrode **160** in the −X direction, and a portion **372** that is linked to the +X direction end portion of the portion **371** and is located below the internal gate electrode **160**.

[0117] Thus, the internal gate electrode **160** is located in the portion of the trench T**1** that is positioned in the termination region. The internal gate electrode **160** may be located in the termination region according to the thickness of the termination region of the insulating film **151** and/or the breakdown voltage of the semiconductor device **100**. This is similar for the other trenches T**2** to T**5** described below as well.

[0118] As shown in FIGS. **10** and **12**, the FP electrode **170** that is in the trench T**2** includes a portion **373** that is next to the −X direction end portion of the internal gate electrode **160** in the −X direction, and a portion **374** that is linked to the +X direction end portion of the portion **373** and is located below the internal gate electrode **160**. The portion **373** is next to the portion **371** of the FP electrode **170** in the trench T**1** in the Y-direction.

[0119] Thus, the internal gate electrode **160** is located also in the portion of the trench T**2** that is positioned in the termination region.

[0120] As shown in FIG. **10**, the arrangement of the FP electrode **170** in the trench T**4** is similar to the arrangement of the FP electrode **170** in the trench T**2**. The arrangements of the FP electrodes **170** in the trenches T**3** and T**5** are similar to the arrangement of the FP electrode **170** in the trench T**1**. Accordingly, a description of the internal configurations of the trenches T**3**, T**4**, and T**5** is omitted.

[0121] Effects of the embodiment will now be described.

[0122] According to the embodiment as well, the pad portion **131** is separated in the +Y direction from the +X direction end portion of the internal gate electrode **160** in each trench T**2**. The distance L between the first covering portion **142***a* and the pad portion **131** in the Y-direction is greater than the distances L**1** and L**2** between the pad portion **131** and the trenches T**2** in the Y-direction. The protrusion **135** protrudes from the pad portion **131**, covers the +X direction end portion of the internal gate electrode **160** in each trench T**2**, and is connected to the +X direction end portion of the internal gate electrode **160** in each trench T**2**. Thus, the internal gate electrode **160** that is in each trench T**2** and is positioned in the Y-direction between the position of the pad portion **131** in the Y-direction and the position of the first covering portion **142***a* in the Y-direction is not easily connected directly to the pad portion **131**. According to the embodiment, such an internal gate electrode **160** can be electrically connected to the external gate electrode **130** by the protrusion **135**. Also, the second covering portion **142***b* covers the contact portion **120***b* of the semiconductor part **120** between the internal gate electrode **160** in the trench T**1** and the internal gate electrode **160** in the trench T**2** positioned furthest toward the −Y side, and is connected to the contact portion **120***b*. Therefore, the effective element area can be increased.

[0123] The first covering portion **142***a* covers the contact portion **120***a* of the semiconductor part **120** adjacent to the internal gate electrode **160** in the trench T**1**, and is connected to the contact portion **120***a*. Therefore, the effective element area can be increased.

[0124] The configurations of the multiple embodiments described above can be combined as appropriate within the extent of technical feasibility. For example, the configuration of the third embodiment may be combined with the second embodiment.

[0125] Embodiments may include the following configurations (e.g., technological proposals).

Configuration 1

[0126] A semiconductor device, comprising:

[0127] a first electrode;

[0128] a semiconductor part located on the first electrode;

[0129] a second electrode located in the semiconductor part, the second electrode extending in a first direction when viewed from above;

[0130] a third electrode located in the semiconductor part, the third electrode extending in the first direction, the third electrode including a first portion, a second portion, and a first middle portion positioned below the second electrode between the first portion and the second portion, the second electrode being located between the first portion and the second portion in the first direction;

[0131] a fourth electrode located above the semiconductor part, the fourth electrode including

[0132] a pad portion separated from the second electrode and the second portion in a second direction crossing the first direction when viewed from above, and

[0133] a protrusion protruding from the pad portion and covering the second electrode, the protrusion being connected to the second electrode;

[0134] a fifth electrode located above the semiconductor part and separated from the fourth electrode, the fifth electrode including

[0135] a first covering portion covering a first contact portion of the semiconductor part adjacent to the second portion when viewed from above, the first covering portion being connected to the first contact portion, and

[0136] a second covering portion covering the first portion, the second covering portion being connected to the first portion; and

[0137] a first insulating film located between the semiconductor part and the second electrode, between the semiconductor part and the third electrode, and between the second electrode and the third electrode.

Configuration 2

[0138] The device according to the configuration 1, wherein

[0139] the first covering portion also covers the second portion, and

[0140] the first covering portion also is connected to the second portion.

Configuration 3

[0141] The device according to the configuration 1 or 2, wherein

[0142] the pad portion includes an end portion in a direction from the second portion toward the first portion, and

[0143] the protrusion protrudes from the end portion of the pad portion.

Configuration 4

[0144] The device according to any one of the configurations 1-3, wherein

[0145] a recess is provided in the fifth electrode,

[0146] the recess is positioned between the first covering portion and the second covering portion, and

[0147] the protrusion is located in the recess.

Configuration 5

[0148] The device according to any one of the configurations 1-4, further comprising:

[0149] a sixth electrode located in the semiconductor part, the sixth electrode being next to the second electrode in the second direction, the sixth electrode extending in the first direction; and

[0150] a seventh electrode located in the semiconductor part, the seventh electrode extending in the first direction, the seventh electrode including a third portion, a fourth portion, and a second middle portion positioned below the sixth electrode between the third portion and the fourth portion, the sixth electrode being located between the third portion and the fourth portion in the first direction; and

[0151] a second insulating film located between the semiconductor part and the sixth electrode, between the semiconductor part and the seventh electrode, and between the sixth electrode and the seventh electrode,

[0152] the protrusion covering the sixth electrode when viewed from above,

[0153] the protrusion being connected to the sixth electrode,

[0154] the third portion being next to the first portion in the second direction,

[0155] the second covering portion also covering the third portion when viewed from above,

[0156] the second covering portion also being connected to the third portion.

Configuration 6

[0157] The device according to the configuration 5, wherein

[0158] the second covering portion also covers a second contact portion of the semiconductor part positioned between the second electrode and the sixth electrode when viewed from above, and

[0159] the second covering portion also is connected to the second contact portion.

Configuration 7

[0160] The device according to the configuration 6, wherein

[0161] the semiconductor part includes:

[0162] a first semiconductor layer of a first conductivity type;

[0163] a second semiconductor layer located in an upper layer portion of the first semiconductor layer, the second semiconductor layer being next to the second electrode with the first insulating film interposed, the second semiconductor layer being of a second conductivity type; and

[0164] a third semiconductor layer located in an upper layer portion of the second semiconductor layer, the third semiconductor layer being of the first conductivity type,

[0165] the third semiconductor layer being located in the second contact portion.

Configuration 8

[0166] The device according to the configuration 7, wherein

[0167] the third semiconductor layer also is located under the first covering portion, and

[0168] the first covering portion also is connected to the third semiconductor layer.

Configuration 9

[0169] A semiconductor device, comprising:

[0170] a first electrode;

[0171] a semiconductor part located on the first electrode;

[0172] a second electrode located in the semiconductor part, the second electrode extending in a first direction when viewed from above;

[0173] a third electrode, the second electrode including a first end portion in an opposite direction of the first direction, the third electrode including a portion next to the first end portion;

[0174] a fourth electrode located in the semiconductor part, the fourth electrode being next to the first end portion of the second electrode in a second direction crossing the first direction when viewed from above, the fourth electrode extending in the first direction, a length in the first direction of the fourth electrode being less than a length in the first direction of the second electrode;

[0175] a fifth electrode, the fourth electrode including a second end portion in the opposite direction of the first direction, the fifth electrode including a portion next to the second end portion in the first direction;

[0176] a sixth electrode located above the semiconductor part, the sixth electrode being connected to the fourth electrode, the sixth electrode including a pad portion and a protrusion, the pad portion being separated from the second and fourth electrodes in the second direction, the protrusion protruding from the pad portion, the fourth electrode including a third end portion in the first direction, the protrusion covering the third end portion, the protrusion being connected to the third end portion; and

[0177] a seventh electrode located above the semiconductor part, the seventh electrode being separated from the pad portion in an opposite direction of the second direction, the seventh electrode including a first covering portion and a second covering portion, a distance from the pad portion to the first covering portion in the second direction being greater than a distance between the fourth electrode and the pad portion in the second direction, the second covering portion covering the portion of the third electrode, the portion of the fifth electrode, and a first contact portion of the semiconductor part, the first contact portion being positioned between the second electrode and the fourth electrode, the second covering portion being connected to the portion of the third electrode, the portion of the fifth electrode, and the first contact portion;

[0178] a first insulating film located between the semiconductor part and the second electrode, between the semiconductor part and the third electrode, and between the second electrode and the third electrode; and

[0179] a second insulating film located between the semiconductor part and the fourth electrode, between the semiconductor part and the fifth electrode, and between the fourth electrode and the fifth electrode.

Configuration 10

[0180] The device according to the configuration 9, wherein

[0181] the first covering portion covers a second contact portion of the semiconductor part,

[0182] the second contact portion is adjacent to the second electrode, and

[0183] the first covering portion is connected to the second contact portion.

Configuration 11

[0184] A semiconductor device, comprising:

[0185] a first electrode;

[0186] a semiconductor part located on the first electrode;

[0187] a second electrode located in the semiconductor part, the second electrode extending in a first direction when viewed from above;

[0188] a third electrode located in the semiconductor part, the third electrode extending in the first direction, the third electrode including a first portion, a second portion, and a first middle portion, the second electrode including a first end portion in an opposite direction of the first direction, the first portion being next to the first end portion in the first direction, the second electrode including a second end portion in the first direction, the second portion being next to the second end portion in the first direction, the first middle portion being positioned below the second electrode between the first portion and the second portion;

[0189] a fourth electrode located in the semiconductor part, the fourth electrode being next to the second electrode in a second direction crossing the first direction when viewed from above, the fourth electrode extending in the first direction;

[0190] a fifth electrode located in the semiconductor part, the fourth electrode including a third end portion in the opposite direction of the first direction, the fifth electrode including a third portion that is next to the third end portion in the first direction;

[0191] a sixth electrode located above the semiconductor part, the sixth electrode including a pad portion and a protrusion, the pad portion being separated in the second direction from the second and fourth electrodes, the protrusion protruding from the pad portion, the fourth electrode including a fourth end portion in the first direction, the protrusion covering the second end portion of the second electrode and the fourth end portion of the fourth electrode, the protrusion being connected to the second and fourth end portions, the protrusion not covering at least a portion of the second portion when viewed from above;

[0192] a seventh electrode located above the semiconductor part and separated from the pad portion in an opposite direction of the second direction, the seventh electrode including a first covering portion and a second covering portion, a distance from the pad portion to the first covering portion in the second direction being greater than a distance between the fourth electrode and the pad portion in the second direction, the second covering portion covering the first portion, the third portion, and a first contact portion of the semiconductor part positioned between the second electrode and the fourth electrode, the second covering portion being connected to the first portion, the third portion, and the first contact portion;

[0193] a first insulating film located between the semiconductor part and the second electrode, between the semiconductor part and the third electrode, and between the second electrode and the third electrode; and

[0194] a second insulating film located between the semiconductor part and the fourth electrode, between the semiconductor part and the fifth electrode, and between the fourth electrode and the fifth electrode.

Configuration 12

[0195] The device according to the configuration 11, wherein

[0196] the first covering portion covers a second contact portion of the semiconductor part,

[0197] the second contact portion is adjacent to the second portion, and

[0198] the first covering portion is connected to the second contact portion.

[0199] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions. Additionally, the embodiments described above can be combined mutually.

What is claimed is:

1. A semiconductor device, comprising:

a first electrode;

a semiconductor part located on the first electrode;

a second electrode located in the semiconductor part, the second electrode extending in a first direction when viewed from above;

a third electrode located in the semiconductor part, the third electrode extending in the first direction, the third electrode including a first portion, a second portion, and a first middle portion positioned below the second electrode between the first portion and the second portion, the second electrode being located between the first portion and the second portion in the first direction;

a fourth electrode located above the semiconductor part, the fourth electrode including

a pad portion separated from the second electrode and the second portion in a second direction crossing the first direction when viewed from above, and

a protrusion protruding from the pad portion and covering the second electrode, the protrusion being connected to the second electrode;

a fifth electrode located above the semiconductor part and separated from the fourth electrode, the fifth electrode including

a first covering portion covering a first contact portion of the semiconductor part adjacent to the second portion when viewed from above, the first covering portion being connected to the first contact portion, and

a second covering portion covering the first portion, the second covering portion being connected to the first portion; and

a first insulating film located between the semiconductor part and the second electrode, between the semiconductor part and the third electrode, and between the second electrode and the third electrode.

2. The device according to claim **1**, wherein

the first covering portion also covers the second portion, and

the first covering portion also is connected to the second portion.

3. The device according to claim **1**, wherein

the pad portion includes an end portion in a direction from the second portion toward the first portion, and

the protrusion protrudes from the end portion of the pad portion.

4. The device according to claim **1**, wherein

a recess is provided in the fifth electrode,

the recess is positioned between the first covering portion and the second covering portion, and

the protrusion is located in the recess.

5. The device according to claim **1**, further comprising:

a sixth electrode located in the semiconductor part, the sixth electrode being next to the second electrode in the second direction, the sixth electrode extending in the first direction; and

a seventh electrode located in the semiconductor part, the seventh electrode extending in the first direction, the seventh electrode including a third portion, a fourth portion, and a second middle portion positioned below the sixth electrode between the third portion and the fourth portion, the sixth electrode being located between the third portion and the fourth portion in the first direction; and

a second insulating film located between the semiconductor part and the sixth electrode, between the semiconductor part and the seventh electrode, and between the sixth electrode and the seventh electrode,

the protrusion covering the sixth electrode when viewed from above,

the protrusion being connected to the sixth electrode,

the third portion being next to the first portion in the second direction,

the second covering portion also covering the third portion when viewed from above,

the second covering portion also being connected to the third portion.

6. The device according to claim **5**, wherein

the second covering portion also covers a second contact portion of the semiconductor part positioned between the second electrode and the sixth electrode when viewed from above, and

the second covering portion also is connected to the second contact portion.

7. The device according to claim **6**, wherein

the semiconductor part includes:

a first semiconductor layer of a first conductivity type;

a second semiconductor layer located in an upper layer portion of the first semiconductor layer, the second semiconductor layer being next to the second electrode with the first insulating film interposed, the second semiconductor layer being of a second conductivity type; and

a third semiconductor layer located in an upper layer portion of the second semiconductor layer, the third semiconductor layer being of the first conductivity type,

the third semiconductor layer being located in the second contact portion.

8. The device according to claim **7**, wherein

the third semiconductor layer also is located under the first covering portion, and

the first covering portion also is connected to the third semiconductor layer.

9. A semiconductor device, comprising:

a first electrode;

a semiconductor part located on the first electrode;

a second electrode located in the semiconductor part, the second electrode extending in a first direction when viewed from above;

a third electrode, the second electrode including a first end portion in an opposite direction of the first direction, the third electrode including a portion next to the first end portion;

a fourth electrode located in the semiconductor part, the fourth electrode being next to the first end portion of the second electrode in a second direction crossing the first direction when viewed from above, the fourth electrode extending in the first direction, a length in the first direction of the fourth electrode being less than a length in the first direction of the second electrode;

a fifth electrode, the fourth electrode including a second end portion in the opposite direction of the first direction, the fifth electrode including a portion next to the second end portion in the first direction;

a sixth electrode located above the semiconductor part, the sixth electrode being connected to the fourth electrode, the sixth electrode including a pad portion and a protrusion, the pad portion being separated from the second and fourth electrodes in the second direction, the protrusion protruding from the pad portion, the fourth electrode including a third end portion in the first direction, the protrusion covering the third end portion, the protrusion being connected to the third end portion; and

a seventh electrode located above the semiconductor part, the seventh electrode being separated from the pad portion in an opposite direction of the second direction, the seventh electrode including a first covering portion and a second covering portion, a distance from the pad portion to the first covering portion in the second direction being greater than a distance between the fourth electrode and the pad portion in the second direction, the second covering portion covering the portion of the third electrode, the portion of the fifth electrode, and a first contact portion of the semiconductor part, the first contact portion being positioned between the second electrode and the fourth electrode, the second covering portion being connected to the portion of the third electrode, the portion of the fifth electrode, and the first contact portion;

a first insulating film located between the semiconductor part and the second electrode, between the semiconductor part and the third electrode, and between the second electrode and the third electrode; and

a second insulating film located between the semiconductor part and the fourth electrode, between the semiconductor part and the fifth electrode, and between the fourth electrode and the fifth electrode.

10. The device according to claim **9**, wherein

the first covering portion covers a second contact portion of the semiconductor part,

the second contact portion is adjacent to the second electrode, and

the first covering portion is connected to the second contact portion.

11. A semiconductor device, comprising:

a first electrode;

a semiconductor part located on the first electrode;

a second electrode located in the semiconductor part, the second electrode extending in a first direction when viewed from above;

a third electrode located in the semiconductor part, the third electrode extending in the first direction, the third electrode including a first portion, a second portion, and a first middle portion, the second electrode including a first end portion in an opposite direction of the first direction, the first portion being next to the first end portion in the first direction, the second electrode including a second end portion in the first direction, the second portion being next to the second end portion in the first direction, the first middle portion being positioned below the second electrode between the first portion and the second portion;

a fourth electrode located in the semiconductor part, the fourth electrode being next to the second electrode in a second direction crossing the first direction when viewed from above, the fourth electrode extending in the first direction;

a fifth electrode located in the semiconductor part, the fourth electrode including a third end portion in the opposite direction of the first direction, the fifth electrode including a third portion that is next to the third end portion in the first direction;

a sixth electrode located above the semiconductor part, the sixth electrode including a pad portion and a protrusion, the pad portion being separated in the second direction from the second and fourth electrodes, the protrusion protruding from the pad portion, the fourth electrode including a fourth end portion in the first direction, the protrusion covering the second end portion of the second electrode and the fourth end portion of the fourth electrode, the protrusion being connected to the second and fourth end portions, the protrusion not covering at least a portion of the second portion when viewed from above;

a seventh electrode located above the semiconductor part and separated from the pad portion in an opposite direction of the second direction, the seventh electrode including a first covering portion and a second covering portion, a distance from the pad portion to the first covering portion in the second direction being greater than a distance between the fourth electrode and the pad portion in the second direction, the second covering portion covering the first portion, the third portion, and a first contact portion of the semiconductor part positioned between the second electrode and the fourth electrode, the second covering portion being connected to the first portion, the third portion, and the first contact portion;

a first insulating film located between the semiconductor part and the second electrode, between the semiconductor part and the third electrode, and between the second electrode and the third electrode; and

a second insulating film located between the semiconductor part and the fourth electrode, between the semiconductor part and the fifth electrode, and between the fourth electrode and the fifth electrode.

12. The device according to claim **11**, wherein

the first covering portion covers a second contact portion of the semiconductor part,

the second contact portion is adjacent to the second portion, and

the first covering portion is connected to the second contact portion.

* * * * *